



The value of relationships in healthcare[☆]

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ABSTRACT

Many interactions between buyers and sellers occur in the context of relationships. In healthcare markets, the patient-primary care physician (PCP) relationship is one common example. I provide evidence that patient-PCP relationships are valued by patients and are good for patients' health. In response to an exogenous loss of a longstanding PCP, adverse events increase: patient mortality increases by 4%, emergency department visits increase by 4%, and hospital admissions increase by 3%. Adverse events increase with the length of the exiting PCP-patient relationship, suggesting that relationships contain non-transferable information that positively impacts patients' health.

1. Introduction

"It is more important to know what sort of person has a disease than to know what sort of disease a person has."—Hippocrates

Standard economic models often assume transactions occur between anonymous buyers and sellers. In practice, however, many exchanges take place in the context of longstanding relationships, such as the client-lawyer, borrower-lender, or patient-primary care physician (PCP) relationship. Of relationships in the healthcare context, a patient's relationship with a PCP may be one of the most important. Worldwide, PCPs form the building block of the healthcare system, coordinating patient care across specialists, catching initial conditions, and referring patients to downstream care (Starfield, 1994). PCP-patient relationships also tend to be long-lasting: 65% of adults have seen their PCP for more than three years (Baker et al., 2003).

Despite their prevalence, it is unclear if patients value relationships and, if so, whether this valuation leads to better health. On the one hand, relationships may be overvalued and detrimental to health outcomes, like doctors performing more C-sections when they know the patient (Johnson et al., 2016). On the other hand, relationships may positively impact patients' health if health-specific information grows over time or if a good match takes time to find (Alsan et al., 2019).

I start by providing descriptive evidence that patients value relationships. I use a travel-cost model to quantify how far patients are willing to travel to maintain their relationship when a PCP moves. I focus on moves where patients can visit moving PCPs by driving ≤ 40 min farther or see replacement PCPs at the original clinic.¹ If patients value the relationship, they should follow moving PCP. If patients view PCPs as interchangeable, patients should see staying PCPs at the original clinic.

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¹ I intentionally focus on cases where patients can maintain care at the original clinic to limit switches where patients may not be able to access care at the original clinic.

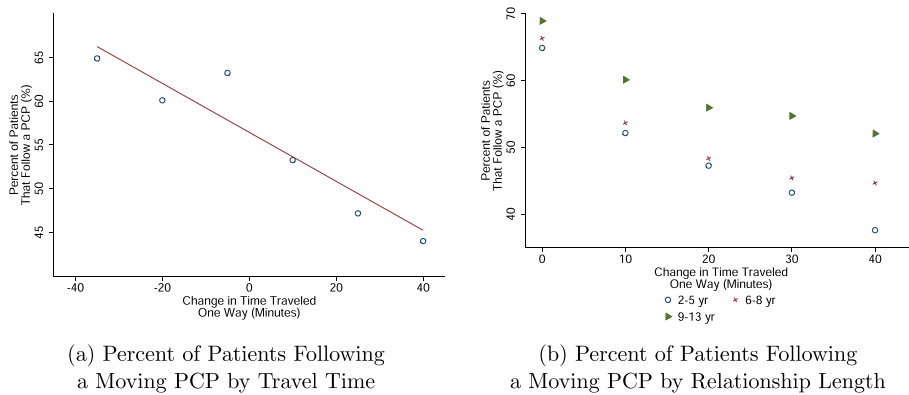


Fig. 1. Patients valuation of PCP relationships using patients' willingness-to-drive. *Notes:* "PCP" stands for primary care physician and "yr" stands for year. The binscatters show the percent of patients following a PCP ≤ 40 minutes from the original clinic, relative to patients' zip code of residence (centroid to centroid). Negative times indicate that the PCP moved closer to the patient's zip code of residence and positive times show that the PCP moved farther. These figures rely on a different sample than the main analyses. Fig. A3a shows the distribution of PCP moves. Fig. 1b drops patients aged < 75 as of $q = -12$ to avoid censoring due to only observing patients once they become Medicare eligible (typically at age 65). Results are similar when expanding the time traveled: see Fig. A3b for moves ≤ 75 min as well as Fig. A3c for results using miles instead of minutes traveled.

I construct a novel dataset quantifying drive time between patient and clinic-physician zip codes in a 20% sample of Medicare data from 2002–2019. I show that, when a PCP moves ≤ 40 min away, 45% of patients follow, costing patients about \$184 additional dollars per year.² Patients with the longest tercile of relationships are 14 percentage points (pp) more likely than patients with the shortest tercile of relationships to follow their moving PCP.³ The evidence suggests that patients value having a relationship with their PCP and that this value increases over the length of the relationship.

I then quantify patients' valuation of relationships in a quasi-experimental framework by leveraging a separate set of moves where patients cannot follow PCPs: physician retirements and relocations > 100 miles away. I quantify the effect in a difference-in-differences research design, using a matched control group of PCPs who did not exit but are otherwise similar to exiting PCPs. For identification, I assume that PCPs depart for idiosyncratic reasons unrelated to the health of their patients 2–3 years before the exit. Outcomes of treated and control patients trend similarly 1–3 years before the exit, supporting this assumption.

The evidence supports a model where patients value relationships, which positively affect patients' health. Relationships determine where patients' demand care: in response to a PCP's exit, only 23% of patients form a new PCP relationship, instead substituting to physicians they know long-term. For instance, patients increase their use of pre-existing specialists by 5.5%. Adverse events also increase in response to the loss of the relationship. Patients are 4% more likely to die, 4% more likely to have an emergency department visit, and 3% more likely to be admitted to the hospital after losing their PCP. Patients are also 4% more likely to be admitted to a skilled nursing facility (SNF) and 3% more likely to initiate hospice and home health care. The effects are consistent across subgroups, such as baseline risk, disability, being a racial minority, and being also eligible for Medicaid ("dual eligibles").⁴

² The number of patients moving with PCPs is large considering that 39 states enforce non-compete agreements (Hausman and Lavetti, 2021). Starr et al. (2021) find that 15–46% of workers were bound by non-competes in 2016, whereas Colvin and Shierholz (2019) show a range of 28–47% in 2019. Zhang et al. (2021) also document that about a third of patients transition with their main PCP when the PCP changes organizations, likely a more severe treatment than the locational moves documented here.

³ 52% of patients with relationships lasting 9–13 years drive ≤ 40 min from patients' zip code of residence to follow a moving PCP, compared to 38% of patients with relationships lasting 2–5 years.

⁴ The percent changes are similar, but the level changes are greater among more vulnerable groups because they use more healthcare at baseline. Similarly, a paper by Staiger (2022) shows that the impact of a PCP's exit among high risk patients is larger in terms of levels but does not report percent changes. For this paper's purposes, I argue that the percent changes are the relevant metric given baseline differences.

To understand why the loss of a PCP impacts patients, I start by ruling out mechanisms not specific to the relationship. I do not find evidence that the firm is disrupted or for network effects by estimating the impact of a PCP's exit on their coworking, staying PCPs and their patients (patients who never saw the exiting PCP).⁵ I find that staying PCPs see 44 more patients per year after a coworking PCP's exit, but the outcomes of staying PCPs' patients are not affected. I also show that access to care does not explain impacts by showing that patients in higher PCP density areas have larger impacts than patients in lower PCP density areas, as opposed to the opposite. Further, I rule out differences between exiting and replacement PCPs rate of referring to specialists as an explanation for patients' increased use of specialists after a PCP's exit.⁶

I then use heterogeneity in the treatment effect to provide evidence in support of relationships positively impacting patients' health. I exploit variation in the length of the PCP-patient relationship to show that adverse events increase with the length of the relationship. The evidence suggests that relationships result from an "experience" component, which increases the value of collaboration over time (Becker, 1962; Jaravel et al., 2018; Polanyi, 1966).

I build on the existing literature by providing evidence that patients value relationships and that relationships positively impact patients' production of health. My findings complement work showing that the loss of a PCP disrupts patient care and has mixed impacts on patient welfare in response to a PCP's retirement among Medicare patients (Sabety et al., 2021; Zhang, 2022; Kwok, 2019), a PCP's exit among high risk Medicaid patients (Staiger, 2022), and a PCP's deployment among military patients (Schwab, 2023); there has also been work in Switzerland (Bischof and Kaiser, 2021) and Denmark (Simonsen et al., 2021) quantifying the impact of practice closures on patients.

I attempt to bring together the current body of work by leveraging the largest sample to-date: PCP moves and retirements in a 20% sample of Medicare patients from 2002–2019. The sample's magnitude gives me precision to quantify rare events like mortality and use heterogeneity to delve into mechanisms. For instance, Sabety et al. (2021) and Zhang (2022) find imprecise impacts on mortality, partly because their samples are 1/5 as large. The long panel also allows me to be the first to examine the relevance of PCP-patient relationships for patient outcomes. I do so by quantifying patients' revealed preference for relationships, that patients substitute to providers they know, and that treatment effects vary by the length of the relationship.

PCP-patient relationships are often severed, whether through

⁵ For instance, Jäger and Heining (2023) shows that worker exits spillover onto co-workers, raising wages and retention probabilities of staying workers for several years.

⁶ This finding relates to differences in practice patterns previously documented by Kwok (2019) and Fadlon and Van Parys (2020).

Table 1

Balance table for exiting PCPs and their patients.

	Treatment	Control	
	Mean (sd)	Mean (sd)	P-values
% Match	75%	47%	
No. of PCPs	19,673	19,673	
No. of Patients	533,446	605,207	
PCP Matching Covariates, 3 Years Before PCP Exit			
PCP Age (yr)	54.7(12.4)	53.8(10.1)	p < 0.001
Caseload per PCP/Month	12.2(10.4)	12.2(10.4)	1.00
Patient Covariates That Were Not Matched On, 3–2 Years Before PCP Exit			
<i>Patient Demographics</i>			
Patient Age (yr)	71.4(6.0)	71.8(5.2)	p < 0.001
White (%)	85.4(22.2)	80.4(26.1)	p < 0.001
Female (%)	63.1(19.8)	63.0(17.4)	0.36
Urban (%)	81.3(39.0)	87.5(33.1)	p < 0.001
<i>Patient Clinical Characteristics</i>			
Elixhauser Risk Score	2.4(0.83)	2.6(0.83)	p < 0.001
End Stage Renal Disease (%)	0.89(3.3)	1.1(3.3)	p < 0.001
Also Enrolled in Medicaid (%)	19.0(22.1)	21.2(24.0)	p < 0.001
<i>Average Annual Rate per Patient</i>			
Primary Care Visits	5.5(2.7)	5.8(2.6)	p < 0.001
PCPs-Clinic Seen	2.5(1.0)	2.0(0.65)	p < 0.001
Specialty Care Visits	9.4(5.0)	9.9(4.8)	p < 0.001
Specialists-Clinic Seen	4.6(2.1)	4.9(2.1)	p < 0.001
Emergency Department Visits	0.68(0.69)	0.65(0.56)	p < 0.001
Inpatient Department Visits	0.34(0.31)	0.35(0.30)	p < 0.001
Annual Patient Expense (\$)	195.76(175.96)	203.84(170.72)	p < 0.001
Annual Spending (\$)	8054.81(5925.10)	8759.98(6441.50)	p < 0.001
Prob. of Death	0.047(0.20)	0.047(0.17)	0.83

Notes: “PCP” stands for primary care physician, “yr” stands for year, “Prob.” stands for probability. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2. PCP age was determined from NPPES, Medicare’s MD-PPAS, and Physician Compare. In addition to the PCP matching covariates reported, gender, PCP training (i.e. MD/DO, PA, NP), and month year of calendar time were exactly matched on. “Caseload per PCP/month” captures the number of patients seen by each PCP for any type of visit, regardless of whether the PCP was assigned as the patient’s PCP. Patient demographics and clinical characteristics are determined from the Medicare Beneficiary Summary File. “Urban” refers to patients’ clinic locations, which were determined using the Center for Medicare and Medicaid Services rural/urban fee schedule. The Elixhauser Risk Score counts comorbidities and pre-existing conditions that are predictive of death; scores range from 0 to 12, with higher scores indicating more coexisting conditions and that patients are “higher risk.” PCPs-Clinic captures the number of PCP-clinics a patient sees. For instance, if they see the same PCP in two different clinics, this will be two. Spending includes provider charges from the carrier file, inpatient charges, and outpatient charges.

insurance network changes, physician retirement, death, and relocation, as well as non-compete agreements that prevent patients from moving with PCPs across firms (Barnett et al., 2017; Sinsky et al., 2022; Hausman and Lavetti, 2021). Results suggest there are trade-offs to artificially severing relationships via contracts like insurance network changes and non-compete agreements. These results are directly relevant to patients, providers, firms, and policymakers attempting to understand and potentially mitigate the harms of disrupting relationships.

The paper proceeds as follows. Section 2 describes how data are constructed. Section 3 quantifies patients willingness-to-follow PCPs. In Section 4, I describe my empirical strategy and identification assumptions. Section 5 presents the aggregate impact of a loss of a PCP. Section 6 explores the specific mechanisms behind the aggregate results. Section 7 concludes.

2. Data construction

2.1. Data sources

I leverage a 20% sample of Medicare patients from 2002–2019 encompassing all healthcare encounters paid by Medicare for about 16 million patients. I observe patients when they become eligible for Medicare (typically age 65) and no longer observe patients when they die or enroll in Medicare Advantage (MA). The data includes a rich set of patient demographics, such as sex, age, race, and zip code of residence. Medicare data also contains about one million healthcare providers and

a nearly nationally representative sample of clinics and providers.⁷

I create a novel linkage to identify clinics, also known as the doctor’s office, by combining the tax identification number (TIN) and nine-digit zip code (ZIP) associated with the billing claim. I restrict the data to only include clinics with under 100 PCPs to overcome the fact that clinics characterized as “large” may actually be multiple clinics.⁸ Further, focusing on clinics with fewer than 100 PCPs increases the likelihood that treated clinics have a common support because large clinics are mechanically more likely to have a departure.⁹ I also exclude non-US clinics which likely have different institutional contexts.

I identify healthcare providers using their National Provider Identifier (NPI), a universal identifier used to submit claims. I supplement the Medicare data with four other data sources by linking across NPIs: the National Plan and Provider Enumeration System, Medicare’s MD-PPAS, Doximity, and Physician Compare. These additional sources allow me to identify providers’ specialty, sub-specialty, sex, age, type of training, and whether NPIs belonged to individuals or organizations.¹⁰ PCPs are

⁷ 93% of American PCPs accept Medicare (Boccuti et al., 2015).

⁸ This excludes large organizations like the Cleveland Clinic, Kaiser, and Intermountain. It is also binding in a handful of cases because the average clinic size is 1.8 PCPs (median 1) as illustrated in Table A3.

⁹ For example, almost all clinics with over 100 PCPs experience a departure within a three year window.

¹⁰ Analyses drop organization NPIs. Table A1 illustrates the algorithm used for each year of data. I use MD-PPAS as my primary data source for PCP age. It only has age starting in 2008, so I extrapolate age backwards, across years for providers I observe over time. I also bring in Doximity to fill in for the handful of PCPs with missing age variables.

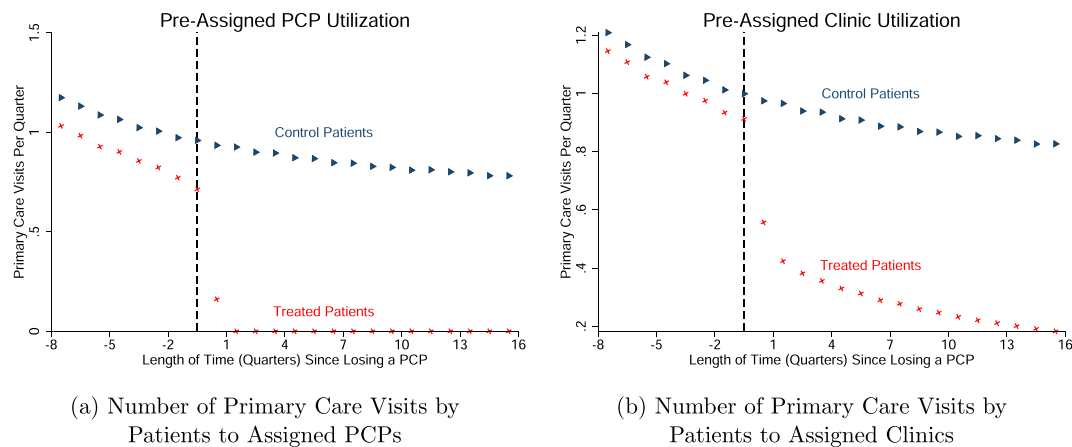


Fig. 2. Number of visits patients make to the assigned PCP and clinic. *Notes:* “PCP” stands for primary care physician. Analyses show raw data using PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. Fig. 2a and Fig. 2b plot patients’ number of primary care visits to assigned (a) PCPs and (b) clinics over relative quarter q . The x-axis ranges from $-8 \leq q \leq 16$. Patients are assigned to PCPs and clinics from $-12 \leq q < -8$ based on where the majority of their primary care was provided. Primary care visits among patients of staying PCPs are reflected in blue triangles and patients of exiting PCPs are reflected in red crosses.

defined as providers with a specialty of family medicine, general practice, geriatric medicine, internal medicine, preventive medicine, pediatrics (many of whom have board certifications in internal medicine), or obstetrics and gynecology (commonly used as PCPs by women, see Leader and Perales (1995)). I follow prior literature and crosswalk between UPINs and NPIs using the NBER’s crosswalk.¹¹

2.2. Primary variable construction

My identification strategy relies on being able to accurately define when a PCP leaves a clinic. I use two separate types of “exits.” First, exits leveraged in Section 3’s descriptive analyses use PCPs that move ≤ 40 minutes from a patient’s zip code of residence, a distance close enough that patients can presumably follow moving PCPs.¹² Second, exits used in the main, quasi-experimental analyses focus on PCPs that patients cannot follow: a move to a new clinic >100 miles away (“relocations”) or cases where PCPs completely exit the data. Complete exits are likely retirements given they occur at an average age of 60 (median 62) (Fig. A1b). The average age of full exit is in line with qualitative work showing that half of physicians 50 years of age and 30% of physicians from age 35 to 49 plan to leave their practice (Gray et al., 2009).

I quantify two categories of outcomes: aggregate markers for poor care and utilization of clinic based services. Markers for poor care include emergency department (ED) visits, inpatient admissions, skilled nursing home (SNF) visits, home health visits, hospice visits, and death. Utilization includes primary and specialty care visits. I determine whether or not the patient is establishing a new PCP relationship by using evaluation and management (E&M) billing codes for office visits. E&M visits include annual exams, wellness visits, physician exams, and consultations. They also flag whether or not the patient is a new patient. In a typical clinic, about 75% of primary care visits are E&M visits. Existing patients make up 96% and new patients make up 4% of primary care E&M visits at baseline. Comparatively, about 24% of specialist visits are for E&M visits.

I define the length of PCP-patient relationships to start from the first time the patient visits the main PCP to the quarter before the PCP’s

departure. I count any visit within a year as a point of contact.¹³ For analyses that leverage the length of PCP-patient relationships I exclude patients under age 75 as of $quarter = -4$ to circumvent left censoring due to only observing most patients when they become Medicare eligible at age 65. (Note: I include ALL Medicare patients—even those less than age 65—in the main analyses.) The sample of patients used to quantify the length of the relationship is well-balanced on observables and is on average 82 years old, or about 15 years older than patients in the main sample (See Table A7).

I categorize the local density of PCPs as the thickness of the local PCP market. It is defined as the number of PCPs filing billing claims within a 30 mile radius of each focal clinic ZIP divided by the total population in that zipcode according to Census estimates.¹⁴ Thick markets are defined to be above average and thin markets are below average density areas.

I use four different patient level definitions: high risk, functional limitations, racial status, and dual eligibles (DEs). I define patients’ risk level using their calculated Elixhauser Index, which creates an index using diagnosis codes that are predictive of death (Elixhauser et al., 1998).¹⁵ I use diagnosis codes three years before the departure to circumvent the potential endogeneity of patient outcomes to treatment. High risk patients have Elixhauser scores in the top quartile of a PCP’s pool of patients. Patients with functional limitations have spinal cord injuries, blindness, mobility impairments, muscular dystrophy, chronic pain fatigue/fibromyalgia, spina bifida, multiple sclerosis, or cystic fibrosis. I use the race variable in the Master Beneficiary Summary File to classify a patient’s race. The DE variable in the Master Beneficiary Summary File classifies whether or not patients are dually eligible for Medicaid.

¹³ Note: this definition is extremely conservative and likely biases estimates to zero. For instance, if a patient saw a PCP once 7 years ago, I would categorize their relationship as lasting 7 years.

¹⁴ I use a 30 mile radius because patients travel 17 miles on average to their assigned clinic, or modal clinic. The distance between five digit zip codes were great-circle distances calculated using the Haversine formula based on internal points in the geographic area. The data set was obtained from the NBER at <https://www.nber.org/data/zip-code-distance-database.html>.

¹⁵ Patients’ Elixhauser scores were derived using the Stata function “Elixhauser.” Patients who had no claims three years before a PCP’s exit were given a risk score of zero.

¹¹ My results are robust to excluding years before 2008, when the UPIN to NPI switch occurred.

¹² I define a move as a change in the PCP’s nine digit zip code and TIN.

Table 2

Treatment effect of a PCP leaving a clinic.

Type	Mean	Impact	Type	Mean	Impact
Utilization of Clinic Based Services			Aggregate Markers for Poor Care		
No. of Specialist and Primary Care Visits	14.8	−0.38*** (0.044) −2.6%	Emergency Department Visits	0.73	0.032** (0.0037) 4.4%
Primary Care Visits	5.0	−0.73** (0.020) −14.8%	Inpatient Visits	0.38	0.011** (0.0025) 3.0%
PCP Visits at Clinic	3.9	−2.2** (0.023) −55.4%	SNF Visits	0.083	0.0019 (0.0012) 2.3%
PCP Visits at Other Clinics	1.029	1.4** (0.019) 140.6%	Home Health and Hospice Visits	0.21	0.0064** (0.0024) 3.0%
Specialist Visits	9.4	0.52** (0.026) 5.5%	Death	0.048	0.0018** (0.00067) 3.7%
Existing SP EM Visits	3.7	0.31** (0.012) 8.3%	Treated PCP Sample Size	19673	
			Control PCP Sample Size	19673	

Notes: “PCP” stands for primary care physician, “SP” stands for specialist, “EM” stands for evaluation and management, “SNF” stands for skilled nursing facility. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4. The sample contains patients of matched PCPs three years before and one year after exit. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Section 2.2 for other variable definitions. Percent changes divide the treatment effect (“impact”) by the baseline control mean (“mean”). See Table A11 for results controlling for linear time trends. See Table A12 for additional outcomes, like the sub-components of the aggregate categories. Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

3. Patients’ valuation of the relationship

I begin by providing descriptive evidence that patients value relationships. I follow Kremer et al. (2011) and use a travel cost model to quantify patients’ revealed preference for relationships. I focus on PCP moves where patients can choose to follow their PCP or continue to seek care with a replacement PCP at the original clinic.¹⁶ The set of PCPs and patients used here are *distinct* from those used in subsequent quasi-experimental analyses.

Fig. A3a shows the distribution of patients average travel time to follow a moving PCP ≤ 75 minutes away. The Figure suggests that mass tapers off around a 40 minutes drive, motivating the use of 40 minutes as a threshold. I count a patient as moving with a PCP if the patient visits the PCP post-move for one or more visits.¹⁷ Table A5 contains sample summary statistics. I use arcGIS to calculate patients centroid to centroid time-traveled (1) from patients’ home to original clinic zip code and (2) the new distance between patients’ home and their PCP’s new clinic. Differencing these two metrics captures how far patients could travel to follow their moving PCP. I then take the average time traveled across a PCP’s patients.

Fig. 1a shows that the percent of patients following moving PCPs is linear in the distance moved. When PCPs move 40 minutes closer to patients, 67.5% of patients move with the PCP. When PCPs move 40 minutes farther, 45% of patients follow.¹⁸ These numbers seem fairly large considering that 39 states enforce non-compete agreements, which legally prohibit PCPs from taking their patients with them when moving practices (Hausman and Lavetti, 2021).

¹⁶ Willingness-to-drive analyses exclude cases where the original clinic also closed to focus on patients valuation of their main PCP versus an alternative PCP at the main clinic.

¹⁷ One or more visit is a relatively weak assignment, so presumably patient movement documented here should be interpreted as a lower bound.

¹⁸ Fig. A3b replicates this analysis up to 75 miles. The pattern is similar but does illustrate a sparser population at farther moves.

I then provide a back-of-the-envelope sense of how much patients are willing to pay to follow known PCPs. To convert minutes to miles, I calculate that the average patient driving 40 minutes is equivalent to 25 miles using the equation from the best fit line in Fig. 1a and Fig. A3c. I then borrow three pieces of information from the literature. First, I use a wage rate of \$13.81 per hour (AARP Public Policy Institute, 2015). Second, 31% of individuals over age 50 have a caregiver who likely accompanies an elder to appointments, especially if traveling a long distance (Reinhard et al., 2019). Third, I assume fuel costs of \$0.13 per mile.¹⁹

The average patient visits their PCP six times per year, which translates to 480 additional minutes, 8 additional hours, or 300 additional miles per year round-trip (25 miles*2*6 visits). At a wage rate of \$13.81, this is the equivalent of \$110.48 in foregone wages. The expected cost of a caregiver’s time is \$34.25, if 31% of individuals have a caregiver who accompanies them. Fuel is \$39. In total, traveling to follow a PCP costs \$184.

Fig. 1b further refines the sample to patients aged ≥ 75 to leverage heterogeneity by terciles of PCP-patient relationship length: 2–5 years, 6–8 years, and 9–13 years. Of patients with 9–13 year long relationships, 52% follow a longstanding PCP that moves 40 minutes away, whereas only 38% of patients with the shortest tercile of relationships do. Patients with the longest tercile of relationship length are therefore 14 percentage points (pp) more likely to follow a PCP who moves ≤ 40 minutes away than patients with the shortest tercile of relationship length.

To assess the robustness of findings, I explore whether or not results are driven by rural areas having larger zip code areas, which could make centroid to centroid driving time look larger. Fig. A3d compares the percent of patients following PCPs ≤ 40 minutes by the concentration of the local PCP market. It shows that patients are 1.7 pp more likely to follow their PCP in low density areas, a small difference that is unlikely

¹⁹ This assumes a cost per gallon of \$3.25 and a vehicle that gets 25 miles per gallon

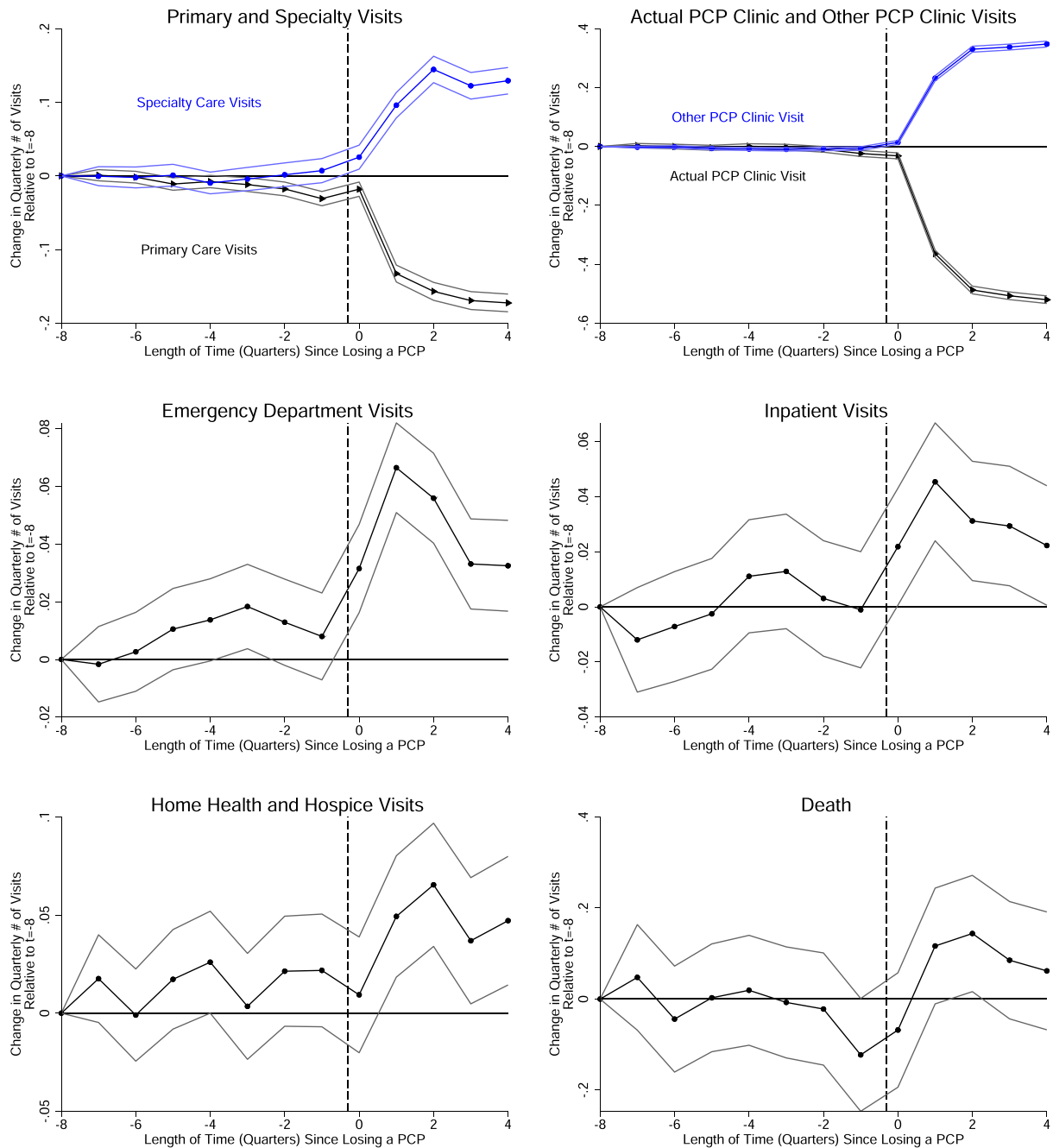


Fig. 3. Impact of a PCP's exit on patients' outcomes. *Notes:* "PCP" stands for primary care physician. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The graph displays results from the difference-in-differences specification outlined in Section 4.4 where regressions are at the PCP-quarter level, contain pre-departure PCP fixed effects, are estimated relative to $q = -8$, and cluster at the PCP-match level. See Table 2 for pooled estimates. See Fig. A5 for substitution from actual PCP clinic to other PCP clinics only focusing on clinics that stay open after a PCP's exit.

to be economically meaningful. Further, the Figure is fairly noisy and the data do not tell a clear story, suggesting that measurement error along this margin is not a main issue. Although minutes traveled is intuitively appealing because it should capture some variability between urban and rural areas that wouldn't be captured with miles, I also replicate analyses using miles instead of minutes traveled (Fig. A3c).

The back-of-the-envelope valuation of \$184 is likely a lower bound. There are undoubtedly additional costs with switching clinics, like moving health records, finding a new clinic, search costs (e.g. due to non-compete agreements), and getting to know new staff, which are not captured here. That being said, the wage rate may be a lower bound on an elderly person's valuation of their time because, by choosing to be

retired, individuals value leisure time at least as much as their wage rate by revealed preference. If one assumed the elder's wage to be zero, the cost of following the PCP is \$73.25, and if one also assumed the elder traveled alone, the cost would only include fuel, \$39. To derive a more accurate willingness-to-pay one would need different prices, which does not apply to the Medicare setting where patients pay the same amount across providers. Precisely estimating patients' willingness-to-pay is a fruitful area for future work.

Patients' Likelihood of Forming a New Relationship

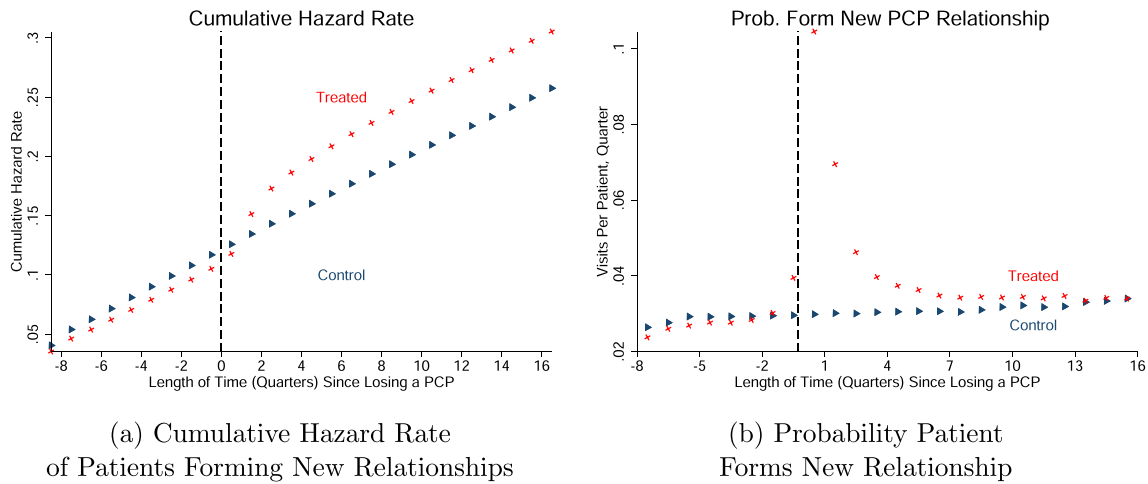


Fig. 4. Effects of a PCP leaving a clinic on patients' likelihood of forming a new relationship. *Notes:* "PCP" stands for primary care physician, "Prob." stands for probability. Analyses show raw data using PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The x-axis contains quarter relative to the departure in $q = 0$. Fig. 4a shows the probability a patient forms a new relationship as a cumulative hazard rate, whereas Fig. 4b shows the probability. Treated patients are represented by red crosses, whereas control patients are represented by blue triangles.

Table 3

Treatment effect of a PCP leaving a clinic by local density of PCPs.

Type	Low PCP Density		High PCP Density		P-Value
	Mean	Impact	Mean	Impact	
Primary Care Visits	5.0	-0.73** (0.020) -14.7%	5.0	-0.81** (0.067) -16.3%	0.22
Prob. Form New PCP Relationship	0.12	0.16** (0.0029) 132.0%	0.11	0.16** (0.0099) 140.9%	0.49
Specialist Visits	9.5	0.51** (0.027) 5.4%	8.5	0.65** (0.098) 7.7%	0.15
ED and Inpatient Visits	0.87	0.036** (0.0043) 4.2%	0.85	0.061** (0.016) 7.2%	0.12
Treated PCPs	18531		1142		
Control PCPs	18653		1020		

Notes: "PCP" stands for primary care physician, "Prob." stands for probability, and "ED" stands for emergency department. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4. The sample contains patients of matched PCPs three years before and one year after exit. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. The regression is at the PCP-year level, contains pre-departure PCP fixed effects, and clusters at the PCP-match level. See Section 2.2 for other variable definitions. Percent changes divide the treatment effect ("impact") by the baseline control mean ("mean"). The local density of PCPs is defined as the number of PCPs filing billing claims within a 30 mile radius of each focal clinic ZIP divided by the population. Thick markets are defined to be above average and thin markets are below average density areas. Fig. A6 contains event plots of main estimates. Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

4. Empirical strategy

4.1. Sample restrictions

I impose four main restrictions. First, I restrict to PCPs who saw more than 30 total patients from $-12 \leq quarters < -8$ to ensure I am not missing departures due to the 20% nature of the Medicare sample.²⁰ This

²⁰ I choose 30 as the cut-off because treatment PCPs see 67 total patients on average (median 26) and control PCPs see 32 total patients on average (median 0) in the *unrestricted* primary care sample. Once I condition on seeing 30 patients, treatment PCPs see on average 130 patients (median 88) and control PCPs see on average 156 patients (median 115) from $-12 \leq quarters < -8$.

sample yields a departure rate of 12% from 2005 to 2018 (Table A2, Appendix A.2). Second, I fix a PCP's pool of patients from $-12 \leq quarters < -8$ before exit, so PCP assignment is plausibly exogenous to the PCP's eventual exit. This circumvents compositional changes in the number and types of patients seen around a PCP's departure which may affect identification. For instance, PCPs may transfer sicker patients to replacement PCPs before healthier patients (or vice versa). Third, I exclude patients not assigned to an exiting PCP (i.e. the exiting PCP is not their modal PCP) but who saw an exiting PCP over the sample period to minimize contamination. Fourth, I exclude co-working PCPs practicing at the same clinic as an exiting PCP from main analyses to ensure that treated PCPs are matched to control PCPs with uncorrelated errors. I bring back in this group of PCPs to measure spillover effects in the clinic level analyses.

Table 4Treatment effect of a PCP leaving a clinic *Controlling for replacement PCP practice patterns.*

	Without Controls		+ Practice Pattern Controls	+ Mortality Controls
Type	Mean	Impact	Impact	Impact
Primary Care Visits	5.0	−0.73** (0.020) −14.6%	−0.71** (0.020) −14.2%	−0.71** (0.020) −14.3%
Specialist Visits	9.5	0.52** (0.026) 5.5%	0.53** (0.026) 5.6%	0.53** (0.026) 5.6%
ED and Inpatient Visits	0.87	0.038** (0.0043) 4.3%	0.038** (0.0042) 4.4%	0.038** (0.0042) 4.4%
Treated PCP Sample Size	19410	19410	19410	
Control PCP Sample Size	19410	19410	19410	

Notes: “PCP” stands for primary care physician. The “without controls” column contains the main specification in Table 2. The leave-one-out average of utilization of replacement PCP’s patients is controlled for non-parametrically using 100 quantiles of the replacement PCP’s non-focal patient’s PCP and specialist utilization in the “+ Practice pattern controls” column. The “+ mortality controls” column adds the leave-out-mean of the PCP’s average mortality rate across patients as a control. Each of the three specifications is estimated in its own regression. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4. The sample contains patients of matched PCPs three years before and one year after exit. PCPs who only saw focal patients (and matched control PCPs) are not included in analyses (263 PCPs). Regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. The without control v. with control results are derived from two separate regressions. Percent changes divide the treatment effect (“impact”) by the baseline control mean (“mean”). See Section 6.1.3 for additional definitions. Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

4.2. PCP matching procedure to select comparison group

My preferred strategy is to coarsen exact match (CEM) observably similar treatment and control PCPs one-to-one, three years before the departure.²¹ I match exactly on month-year of calendar time to derive a relative time measure for control PCPs and their associated patients, PCP sex, and the type of PCP (i.e. NP, PA, or MD). I also match on four coarsened bins of PCP age in $quarter = -12$ and 10 coarsened bins of the number of patients seen from $-12 \leq quarter < -8$.

I then assign patients to PCPs based on their modal number of E&M visits $-12 \leq quarters < -8$ before the departure. I use E&M visits for the assignment in order to isolate visits made to a patient’s PCP as opposed to secondary staff.^{22,23} A similar procedure was used to define the clinic-level sample detailed in Appendix B.

Table A3 compares the full sample to the matched sample, showing that groups are similar. I match for at least four reasons. First, departing PCPs are older than staying PCPs – see Fig. A1a for the age distribution of staying vs. departing PCPs. Older PCPs may be different from younger PCPs in ways that affect their patients outcomes; they also have older patients. Second, the rate of departure as well as practice styles may differ by a PCP’s gender and training (e.g. nurse practitioner (NP), physician assistant (PA), or medical doctor (MD)). Third, the physician workforce is in the midst of a burnout crisis and a growing number of PCPs are leaving clinical practice due to large and growing caseloads (Sinsky et al., 2022; Sabety et al., 2021). High caseloads are therefore potentially correlated with patient outcomes — e.g. more burdened PCPs may have less time for individual patients. As a result, controlling for these dimensions are likely important when comparing treated to control units. Fourth, matching assigns control units a synthetic match event date, which enables visual assessment of pre-trends in the raw data. It also stacks the data, avoiding issues documented in a large and growing literature on common pitfalls when using difference-in-

differences designs (Roth and Rambachan, 2022; Sun and Abraham, 2021).

4.3. Summary statistics

Table 1 describes summary statistics for PCPs and patients by treatment status. Comparing treatment and control group means, standard deviations, and p-values shows that there are small differences between groups, but they do not appear to be economically meaningful.²⁴

Match Rate. The first section of Table 1 illustrates the match rate and resultant sample size. Of treated PCPs that meet the sample restrictions, 75% are matched, leading to a sample of 19,673 control and 19,673 treatment PCPs. This translates to 533,446 treated patients and 605,207 control patients. The strongest restriction is exactly matching on whether the PCP is a MD, NP, or PA (Table A4).

PCP Matching Covariates. The second panel shows the matching covariates defining the coarsened bins. Treated PCPs are about one year older than control PCPs and treated and control PCPs have similar caseloads.

Patient Covariates by Treatment Status. The third section of the table shows patient covariates. The count of PCP visits are about half of specialty visits. PCPs-Clinic capture the count of PCP clinic units. For instance, if a patient sees the same PCP but at different clinics, this will be considered two. On average, 3–2 years before PCP exit patients see 1.5 different PCPs across clinics. Of primary care visits, 90% occur at the assigned clinic and 86% of their pre-period visits are with the main PCP at the assigned clinic. This implies that the .5 additional PCPs patients see on average are at a different clinic.

Treated and control patient characteristics also appear broadly similar. If anything, the characteristics trend towards control patients being in worse health than treated patients, which could potentially bias results to the null.

4.4. Estimating equations and identification

I estimate the causal impact of a PCP’s departure on patient

²¹ I use coarsened exact matching following recent literature (e.g. Jäger and Heining (2023); Jaravel et al. (2018); Sarsons (2017); Azoulay (2010)). See Table A4 for the relative weight of each restriction.

²² Section 2.2 provides additional details on E&M visits.

²³ This allocation method follows the standard in the literature (e.g. Pollack et al. (2016)). If a patient is assigned to a PCP that is not in the matched sample, that patient and matched PCP-patient pair is excluded from analyses.

²⁴ Normalized differences over 0.30 are considered large and economically significant (Imbens, 2015).

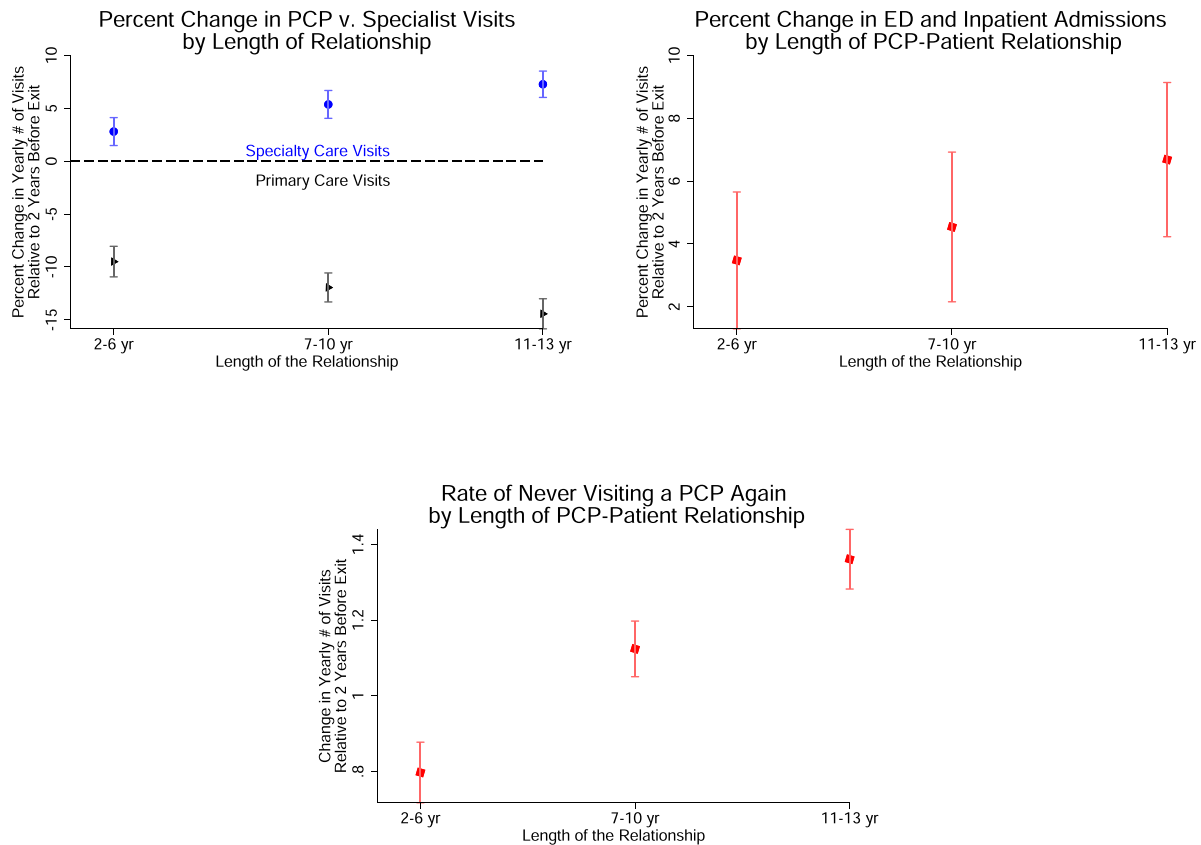


Fig. 5. Impact of exit on adverse events by length of exiting PCP-patient relationship. *Notes:* “PCP” stands for primary care physician, “ED” stands for emergency department, and “yr” stands for year. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. Patients are restricted to be over age 75. The graphs plot each pooled coefficient from the difference-in-differences specification outlined in Section 4.4 one year post-departure. The regression is estimated at the PCP-year level, contains pre-departure PCP fixed effects, clusters at the PCP-match level, and is estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. See Section 2.2 for more details on how heterogeneity and variables were defined. See Table 5 for pooled point estimates across outcomes. See Fig. A4 for standard event-study graphs with the pre-period. See Table A7 for balance across groups. See Fig. A8 for impacts four years. after PCP exit.

outcomes using an event study, difference-in-differences design over a stacked sample that matches PCPs, as outlined in Section 4.1 and 4.2. Regression results rely on equations of the form:

$$y_{ji} = \sum_{\tau=-2}^1 \beta_{\tau}^{Treated} \times \mathbf{1}(i = \tau) \times Treated_j + \sum_{\tau=-2}^1 \beta_{\tau} \times \mathbf{1}(i = \tau) + \rho_j + \epsilon_{ji} \quad (1)$$

where y_{ji} denotes the average outcomes over PCP j 's pool of patients in relative time i . The definition of relative time is the difference between calendar time and when the PCP exits. $\mathbf{1}(i = \tau)$ are relative time fixed effects and ρ_j are PCP fixed effects, which absorb average differences across PCPs.²⁵ I cluster standard errors at the pre-departure PCP-match level to account for idiosyncratic factors specific to a matched pair. This assumes that errors of matched pairs are uncorrelated.^{26,27}

²⁵ As a robustness check, I estimate main results with PCP-match fixed effects and estimate the main specification clustering at the firm level. Both yield similar results.

²⁶ I restrict the data to support the plausibility of this assumption and do not allow PCPs practicing at the same clinic as an exiting PCP to enter the main set of results as outlined in Section 4.1. However, I do leverage these PCPs in Section 6.1.1 when testing for spillover effects.

²⁷ I start at the patient level and sum over patient p 's y_{pji} s within a PCP's pool of patients. I then normalize this outcome by the number of patients in each PCP's pool, summing over the monthly PCP level averages of each outcome. To obtain estimates representative of the original patient population, frequency weights are used in all regressions.

Identification relies on comparing outcomes within a PCP's group of patients to the matched control PCP's pool of patients, relative to the omitted time group. Heterogeneity analyses incorporate interaction terms into the regression framework to identify separate treatment effects while simultaneously controlling for time invariant factors.

I use three time units throughout the paper: year, quarter, and month. The pooled regressions are at the relative year level. Most of the visual event plots are at the relative quarter level. Clinic level results use the month level to pick up fine-grained changes. Year level analyses estimate regressions relative to $year = -2$ and quarter, $quarter = -8$, to allow for anticipatory effects without biasing main estimates. (There are no regressions using the month level.)

5. Aggregate impact of the loss of a PCP on patients

I now turn to results from the quasi-experimental framework outlined in Section 4 to understand how patients' valuation of relationships affects their health. I leverage PCP retirements and relocations > 100 miles away, where patients are unable to follow the moving PCP. Fig. 2a and 2b illustrate the identifying variation by plotting the average number of visits to assigned PCPs and clinics over relative quarter q for 16 quarters, or four years, after a PCP's exit. For instance, if the average number of primary care visits per quarter is one, patients visit their main PCP around four times a year. Blue triangles represent control patients and red crosses represent treated patients. Patients are assigned to PCPs from $-12 \leq q < -8$ and $q = 0$ marks the last quarter exiting PCPs see patients. Treated and control patients do not visit their main PCP at the

Table 5

Treatment effect of a PCP leaving a clinic by relationship length.

Type	2–8 Year		9–13 Year	
	Mean	Impact	Mean	Impact
Primary Care Visits	4.9	−0.52*** (0.030) −10.5%	5.7	−0.79*** (0.035) −13.9%
Patients that Formed a New PCP Relationship	0.094	0.12*** (0.0033) 123.7%	0.11	0.20*** (0.0050) 180.4%
Specialist Visits	8.9	0.30*** (0.048) 3.4%	9.9	0.67*** (0.054) 6.7%
Emergency Department Visits	0.73	0.026*** (0.0069) 3.6%	0.70	0.041*** (0.0076) 5.8%
ED Visits, Primary Care Treatable	0.48	0.015*** (0.0053) 3.1%	0.45	0.026*** (0.0056) 5.7%
Inpatient Visits	0.45	0.0083 (0.0052) 1.8%	0.40	0.017*** (0.0056) 4.2%
Prob. of Death	0.092	0.0046** (0.0017) 5.0%	0.058	0.0040** (0.0016) 7.0%
Treated PCP Sample Size	16865		9984	
Control PCP Sample Size	17204		11301	

Notes: “PCP” stands for primary care physician, “ED” stands for emergency department, “Prob.” stands for probability. Analyses use PCP moves of >100 miles and complete exits; patients are over age 75. The Table displays results from interaction terms from the difference-in-differences specification outlined in Section 4.4 over the full, matched sample shown in Table 1. The sample contains patients of the main sample of matched PCPs three years before and one year after exit, as shown in Table 1. The only restriction on this sample of PCPs is that they have to see at least one patient over age 75, which drops 2,124 PCPs. The regression is estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the patient-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Table A7 for balance across groups. Percent changes divide the treatment effect (“impact”) by the baseline control mean (“mean”). Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

Table A1

Prioritization of data sources.

Outcome	1 st	2 nd	3 rd	4 th
NP, PA, or MD/DO	NPES	MD-PPAS	PC	Claims
Physician Specialty	MD-PPAS	Claims		
Age	MD-PPAS	Doximity		
Gender	Modal Response			

Notes: The National Plan and Provider Enumeration System (NPES), Medicare’s MD-PPAS, Physician Compare (PC), Doximity, and Medicare’s Part B Carrier file (Claims) were combined and prioritized based on reliability to determine the type of provider, specialty, sub-specialty age and gender based on health providers’ national provider identifiers (NPI). MD/DO, PA, and NP stands for medical doctors/doctors of osteopathy, physician assistants, and nurse practitioners, respectively. “Modal Response” means that the modal gender across all five data sources was considered the correct sex.

same rate because treated patients see slightly more PCPs than control patients at baseline (Table 1).

As expected, Fig. 2a shows that patients no longer see exiting PCPs after a PCP exits.²⁸ Treated and control patients see assigned PCPs at the same rate before a PCP’s exit, as illustrated by the curves moving in parallel. Both curves slope downward due to mean reversion and patients dying over time. Curves begin to separate the quarter before the departure implying that exiting PCPs may see fewer patients leading up to the departure. This motivates estimating event studies relative to $q = -8$ and year-level regressions relative to 2–1 years before exit to allow for anticipation. Fig. 2a shows that visits decrease from 0.71 visits at $q = -1$, to 0.16 visits at $q = 0$, to 0 at $q = 1$ in response to a PCP’s exit.

Fig. 2b graphs patients’ visits to the main clinic over relative quarter q . The graph is similar to Fig. 2a except slightly closer in terms of levels. In response to a PCP’s departure, patients’ have 0.91 fewer visits to the

main clinic in $q = -1$ and 0.55 fewer visits in $q = 0$.

5.1. How the loss of a PCP affects patient outcomes

This section documents that relationships determine where patients demand care and positively impact patients’ health. Table 2 shows point estimates and Fig. 3 shows event plots for main outcomes. Fig. 3 shows minor pre-trending and anticipation for primary care visits, which motivates the use of a regression specification where all estimates are relative to $-8 \leq q < -4$. I also document main effects controlling for linear time trends in Table A11, showing that effects are maintained, further supporting the regression strategy.

I start by documenting the aggregate effects. In response to a PCP’s exit, total visits (specialist + PCP) decrease by −0.21 visits per year, a −1.5% decrease (Table 2). Decomposing the nearly unchanged total visit count shows that patients substitute from PCPs to specialists. Primary care visits decrease by −0.73 visits per year and specialist visits increase by 0.52 visits per year (see Fig. 3 for event plots). I also find that patients are more likely to substitute to specialists who act closest to

²⁸ Visits are not zero in $q = 0$ because PCPs stop seeing patients at various times during the quarter.

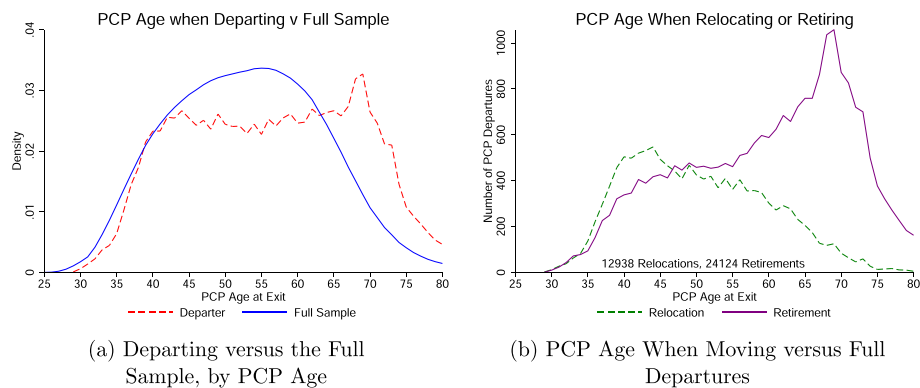


Fig. A1. Histograms of PCP age. *Notes:* The Figures count each departing PCP once. For example, if a PCP departed from a clinic multiple times, they will be counted once, in the first year the PCP departs..

Table A2
Breakdown of departures.

	All Departures	Full Departures	Moves
2005	386	237	149
2006	680	258	422
2007	1,278	659	619
2008	1,611	893	718
2009	1,312	756	556
2010	1,728	1,061	667
2011	1,950	1,216	734
2012	2,222	1,419	803
2013	2,333	1,569	764
2014	2,600	1,795	805
2015	2,543	1,824	719
2016	2,549	1,808	741
2017	2,730	2,054	676
2018	3,074	2,368	706
Total:	26,996	17,917	9,079

Notes: The Table counts the total number of departures per year. If a PCP departed from a clinic multiple times, they will be counted each time, so long as departures occurred in different years.

Table A3
Comparison across full and matched sample.

	Full Sample	Matched Sample
No. of PCPs	69,977	39,346
Treated	26,996	19,673
Full Departures	17,917 (26%)	12,884 (33%)
Moving Sites	9,079 (13%)	6,789 (17%)
PCP Age	53.40	54.64
Caseload per PCP/month	17.52	16.06
Number of PCPs at Practice	1.88	1.79
Urban Location (%)	86.72	87.58
Female (%)	24.36	27.75

Notes: The Table shows characteristics of the full sample against the matched sample shown in Table 1.

PCPs—e.g. nephrologists, cardiologists, and gastroenterologists—opposed to specialists that deliver short-term, condition specific care, such as surgeons (Table A12). Patients' substitution away from PCPs to specialists is maintained for at least four years after a PCP's exit (Table A13). Table A21 partially explains this fact by showing that patients who never visit a PCP again drive the aggregate decline in primary care visits; whereas, patients who visit another PCP after the main PCP's exit seem to utilize a similar amount of primary care post-exit.

I leverage the richness of the data to provide evidence that the loss of the relationship explains impacts. I show that patients substitute from the main PCP clinic to other PCP clinics, even in cases where the main

Table A4
Match Attrition for Each Matching Variable

Dropping	Treatment Match Rate
—	87%
PCP Age	89%
Female	88%
Patients Seen at t=24	92%
NP, PA, or MD	94%

Notes: The Table shows the match rate for the treated sample in Table 1.

clinic remains open (see Fig. 3 for all cases and Fig. A5 for only open clinics).²⁹ Further, although all treated patients lose a PCP, only 23% visit a new PCP and 27% of patients never visit a PCP again, choosing instead to visit specialists they know more frequently. The remaining 50% of patients substitute to a PCP they have known for on average 5.7 years, 1.7 years shorter than the 7.4 years this subset of patients knew their exiting PCP.³⁰

To provide more context for the 50% of patients switching to PCPs they know, I document several descriptive facts on the difference between exiting and known replacement PCPs showing they're fairly similar. The average exiting PCP has 21.8 years of experience, whereas the replacement PCP has 20.9 years of experience. In terms of training, 85% of patients switch from a MD to MD, 10% switch from an MD to NP or PA, and 3% of patients switch from an NP or PA to MD. The average exiting PCP is 57.7 years old and the average replacement PCP is 56.2 years old.

I then show that the loss of a longstanding PCP relationship negatively impacts patients' health. Most importantly, patients who lose a longstanding PCP are 3.7% more likely to die (Table 2 and Fig. 3). I support this finding by showing that ED visits increase 4.4% per year, inpatient visits increase 3.0% per year, skilled nursing facility visits increase 4.1% per year, and home health and hospice visits increase 3.4% per year (Table 2).

Table A14 shows these effects are consistent across subgroups, such as risk, disability, dual eligible (DE), and whether or not the patient is a racial minority, further confirming the main effects documented in Table 2. One potential explanation for findings could also be that it is hard to find another PCP. While Section 6.2 explores this in more depth, Table A15 shows these effects are mirrored in the subset of clinics that stay open—where patients should be able to easily shift care to a replacement PCP after a PCP's exit. Table A12 provides additional

²⁹ This provides further evidence against the hypothesis that patients are unable to access care.

³⁰ For the cases where a patient sees multiple PCPs they know post departure, I report the maximum relationship between the replacement PCP options. For other characteristics, like years practicing, I take the average.

Table A5Balance Table for PCPs who move ≤ 40 minutes away.

	All Patients	2–5 Year Relationship	6–8 Year Relationship	9–13 Year Relationship
No. of PCPs	4,715	3,373	3,338	2,194
PCP Age	50.5	47.5	49.5	54.7
Caseload per PCP/month	21.7	19.5	22.1	25.3
Urban Location (%)	88.9	89.5	89.0	90.4
Female (%)	26.9	31.9	26.4	20.3
No. of Patients	115,502	12,242	16,606	18,811

Notes: The PCP sample includes PCPs who move ≤ 40 min from specific patients. The “all” column contains all patients within 40 min of the moving PCP’s new clinic, whereas the relationship length columns also restrict to patients ≥ 75 years of age. PCP age was determined from NPPES, Medicare’s MD-PPAS, Doximity, and Physician Compare. PCPs are specific to the clinic and are defined as a NPI, tax number, zipcode unit. [Table A18](#) shows outcomes among the sample of patients with a PCP who moves ≤ 40 min away. [Table A6](#) shows covariates among only patients.

Table A6Covariates of Patients with PCPs moving ≤ 40 minutes away.

	All Patients	2–5 Year Relationship	6–8 Year Relationship	9–13 Year Relationship
Patient Age (yr)	72.0	80.9	80.7	80.9
White (%)	88.9	91.0	91.2	91.5
Female (%)	61.4	67.2	65.4	63.4
Elixhauser Risk Score	245.2	260.9	263.6	281.5
End Stage Renal Disease (%)	0.43	0.11	0.13	0.19
Also Enrolled in Medicaid (%)	11.8	8.3	7.8	5.9

Notes: The PCP sample includes patients whose PCPs who move ≤ 40 min away.

context for main results, by quantifying the impact on sub-categories of ED utilization, timing of visits, and spending. Taken together, results strongly suggest that relationships determine where patients demand care and positively impact patients’ health.

I further reinforce the evidence by focusing on the separate sample of PCPs and patients used in Section 3, where PCPs move ≤ 40 minutes away from patients, a distance close enough for patients to hypothetically follow PCPs.³¹ I derive a matched sample of control PCPs and patients following the methods in Section 4.4. If maintaining a PCP relationship is good for patients health, patients who can follow a moving PCP should not experience adverse events in response to a PCP’s move. [Table A18](#) shows precisely this: adverse events are eliminated when patients have the option to follow their PCP.

The evidence builds on our existing understanding of how patients are impacted by a PCP’s exit. For instance, [Sabety et al. \(2021\)](#) and [Zhang \(2022\)](#) look specifically at PCP retirements showing that patients decrease their use of primary care by 9–18%, and increase their use of specialists by 6–8% on aggregate.³² Neither paper focuses on relationships as a main mechanism or detects a significant effect on mortality, in part because their samples are 1/5 of the size of the sample used here.

In contrast, I use a sample of PCP exits that include both retirements and moves, giving me power to quantify the impact of a PCP’s exit on mortality and allowing me to focus on a slightly different question in a more representative sample (and likely explaining why the effect sizes

are not the same).³³ For instance, [Zhang \(2022\)](#) finds that a PCP’s retirement insignificantly increases mortality by 0.02% (standard error 0.02), a 1% increase off a mean mortality rate of 1.75%. The small average mortality rate of 1.75% suggests that [Zhang \(2022\)](#)’s analyses use a healthier subsample because the average Medicare beneficiary has a mortality rate between 5.35% and 4.37% from 1999–2017 ([Newhouse et al., 2019](#)).³⁴ Similar to [Newhouse et al. \(2019\)](#), I find an average mortality rate of 4.8% across years, suggesting that the sample used in this paper is more representative of average Medicare patients.

5.2. Robustness checks

I address three main concerns. First, patients may switch to MA in response to a PCP’s exit, which would create a missing data problem because MA patients are unobserved. This could occur if I misclassify PCPs who stop taking Traditional Medicare (TM) as “exiting,” and patients switch to MA to continue seeing their PCP. Somewhat alleviating this concern, [Table A12](#) shows that patients do not systematically switch into MA in response to a PCP’s exit.

Second, I show that results are robust to changing the match specification. The results in [Table A17](#) modify the match to include whether the PCP exited from an urban or rural area. The results largely confirm the main findings in [Table 2](#).

Third, It could also be the case that clinic-level changes are

³¹ As noted in Section 3, among PCPs who move 40 minutes closer to the patient, 67.5% of patients move with the PCP. When PCPs move 40 minutes farther, 45% of patients choose to follow.

³² [Zhang \(2022\)](#) has a different empirical approach, where she restricts the sample to patients who visited a PCP from 2006–2007 and follows this sample through 2013. Treated patients lost the PCP they saw in 2006/2007 from 2010 and 2013, whereas control patients are those whose 2006/2007 PCP did not retire by 2013. For a PCP to retire, they have to be aged 60 or older.

³³ [Table A16](#) shows main effects restricting to only PCP exits that are full departures. This definition is much closer to “retirements” used in [Zhang \(2022\)](#) and is exactly the same as the exit definition used in [Sabety et al. \(2021\)](#). The results are similar, but less well powered.

³⁴ In terms of additional covariates, the sample in this paper appears more nationally representative than [Zhang \(2022\)](#). For instance, patients included in [Zhang \(2022\)](#) are on average 76 years old in the baseline 2006 period and 12% are dually eligible. Comparatively, patients are on average 71 years old in year $t-2$ and 20% are dually eligible for Medicaid in this paper. Nationally, the average beneficiary age is between 65 and 74 and 17% of TM patients are dually eligible, accounting for 33% of spending ([Tarazi et al., 2022](#); [Peña et al., 2023](#)).

Table A7

Balance table for PCPs and patients by length of relationship.

	Only Treated PCPs and Patients		P-Values
	2–8 Yrs	9–13 Yr	
Sample Sizes			
No. of PCPs	16,172	13,592	
No. of Patients	94,637	145,897	
PCP Matching Covariates 3 Years Prior to PCP Exit			
Avg PCP Age (yr)	53.8	57.9	p < 0.001
Caseload per PCP/Month	13.1	14.2	p < 0.001
Outcomes That Were Not Matched On			
<i>Patient Demographics</i>			
Patient Age (yr)	82.1	81.9	p < 0.001
PCP-Patient Bond (yr)	4.6	10.4	p < 0.001
White (%)	90.9	91.5	p < 0.001
Female (%)	68.1	62.3	p < 0.001
Urban (%)	77.9	80.2	p < 0.001
<i>Patient Clinical Characteristics</i>			
Elixhauser Score	2.7	2.8	p < 0.001
Also Enrolled in Medicaid (%)	11.5	8.2	p < 0.001
<i>Average Annual Rate per Patient, 2–3 Years Before Exit</i>			
Primary Care Visits	5.4	6.1	p < 0.001
PCPs Seen	2.3	2.4	0.004
Specialty Care Visits	9.5	10.0	p < 0.001
Specialists Seen	4.1	5.1	p < 0.001
Emergency Department Visits	0.72	0.62	p < 0.001
Inpatient Department Visits	0.42	0.35	p < 0.001
Annual Patient Expense (\$)	873.45	659.42	p < 0.001
Annual Spending (\$)	4142.04	3934.83	p < 0.001
Prob. of Death	0.12	0.043	p < 0.001

Notes: Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table and related analyses only include patients ≥75 years of age. The underlying sample follows treatment and control patients 1 year after a PCP's exit. See Table 1 for a further description of outcomes. Values are as of 3 years before a PCP's exit, unless otherwise noted. See Table A8 and Table A9 for balance tables specific to relationship bins.

contaminating the PCP treatment effect. To test for this, Fig. A2 plots the number of PCPs exiting over time. It shows that subsequent PCP exits do not systematically happen at the treatment threshold, but rather the line trends smoothly downward over time. Therefore, there do not appear to be systematic changes occurring at the clinic that are driving effects.

6. What is the right model of relationships?

A key finding is that relationships determine where patients demand care. Patients travel with moving providers, and, in the quasi-experimental framework, patients substitute to providers they know after losing a longstanding PCP relationship. Documenting these facts adds significantly to our understanding of how the loss of a longstanding relationship impacts patients.

This Section attempts to add to our understanding of mechanisms. I do not claim that any one exercise is entirely conclusive. Instead, through this series of exercises, I provide suggestive evidence that the loss of information specific to the relationship explains impacts. I do this in two steps. First, I rule out alternative mechanisms not specific to the relationship. Second, I show evidence supporting the relevance of relationships to patients' health. With the caveat that these results are not causal, I then empirically show the evidence more closely aligns with the experience, as opposed to the match quality, view of relationships.

6.1. Ruling out mechanisms not specific to the relationship

6.1.1. Ruling out patients not being able to access care at main clinics

It could be the case that patients substitute to non-primary care settings because they are unable to access care at main clinics. I test this by quantifying the impact of a PCP's exit on staying, co-working PCPs

Table A8

Balance table for PCPs and patients by 2–8 year long relationships.

	Treated	Control	P-Values
Sample Sizes			
No. of PCPs	16,172	17,000	
No. of Patients	94,637	99,688	
PCP Matching Covariates 3 Years Prior to PCP Exit			
Avg PCP Age (yr)	53.8	54.1	0.02
Caseload per PCP/Month	13.1	12.9	0.05
Outcomes That Were Not Matched On			
<i>Patient Demographics</i>			
Patient Age (yr)	82.1	82.1	0.72
PCP-Patient Bond (yr)	4.6	4.6	p < 0.001
White (%)	90.9	86.2	p < 0.001
Female (%)	68.1	68.7	0.004
Urban (%)	77.9	86.3	p < 0.001
<i>Patient Clinical Characteristics</i>			
Elixhauser Score	2.7	2.9	p < 0.001
Also Enrolled in Medicaid (%)	11.5	14.5	p < 0.001
<i>Average Annual Rate per Patient, 2–3 Years Before Exit</i>			
Primary Care Visits	5.4	5.6	p < 0.001
PCPs Seen	2.3	1.9	p < 0.001
Specialty Care Visits	9.5	10.0	p < 0.001
Specialists Seen	4.1	4.4	p < 0.001
Emergency Department Visits	0.72	0.73	0.28
Inpatient Department Visits	0.42	0.46	p < 0.001
Annual Patient Expense (\$)	873.45	1022.27	p < 0.001
Annual Spending (\$)	4142.04	4605.53	p < 0.001
Prob. of Death	0.12	0.13	0.27

Notes: Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table and related analyses only include patients ≥75 years of age. The underlying sample follows treatment and control patients 1 year after a PCP's exit. Values are the mean. See Table 1 for a further description of outcomes. Values are as of 3 years before a PCP's exit, unless otherwise noted.

Table A9

Balance table for PCPs and Patients by 9–13 year long relationships.

	Treated	Control	P-Values
Sample Sizes			
No. of PCPs	13,592	15,302	
No. of Patients	145,897	171,498	
PCP Matching Covariates 3 Years Prior to PCP Exit			
Avg PCP Age (yr)	57.9	56.4	p < 0.001
Caseload per PCP/Month	14.2	13.6	p < 0.001
Outcomes That Were Not Matched On			
<i>Patient Demographics</i>			
Patient Age (yr)	81.9	81.9	0.001
PCP-Patient Bond (yr)	10.4	10.4	p < 0.001
White (%)	91.5	86.3	p < 0.001
Female (%)	62.3	62.9	p < 0.001
Urban (%)	80.2	86.0	p < 0.001
<i>Patient Clinical Characteristics</i>			
Elixhauser Score	2.8	3.0	p < 0.001
Also Enrolled in Medicaid (%)	8.2	12.9	p < 0.001
<i>Average Annual Rate per Patient, 2–3 Years Before Exit</i>			
Primary Care Visits	6.1	6.2	p < 0.001
PCPs Seen	2.4	1.8	p < 0.001
Specialty Care Visits	10.0	10.4	p < 0.001
Specialists Seen	5.1	5.3	p < 0.001
Emergency Department Visits	0.62	0.60	0.01
Inpatient Department Visits	0.35	0.36	0.14
Annual Patient Expense (\$)	659.42	730.00	p < 0.001
Annual Spending (\$)	3934.83	4315.65	p < 0.001
Prob. of Death	0.043	0.048	0.24

Notes: Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table and related analyses only include patients ≥75 years of age. The underlying sample follows treatment and control patients 1 year after a PCP's exit. Values are the mean. See Table 1 for a further description of outcomes. Values are as of 3 years before a PCP's exit, unless otherwise noted.

Table A10

Shift of care from primary care to specialty setting.

PCP Administered Type	Mean	Impact	Specialist Administered Type	Mean	Impact
Medications					
Filled Prescriptions from PCP	10.9	−0.92** (0.054) −8.5%	Filled Prescriptions from Specialists	5.5	0.75** (0.037) 13.8%
Chronic Medication Fills from PCP	4.5	−0.32** (0.022) −7.2%	Chronic Medication Fills from Specialists	1.6	0.31** (0.015) 19.2%
Preventive Care					
Preventive Care by PCP	1.2	−0.28** (0.010) −23.3%	Preventive Care by SP	0.32	0.099** (0.0036) 31.1%
Flu Vaccine by PCP	0.26	−0.063** (0.0022) −24.1%	Flu Vaccine by Specialist	0.038	0.018** (0.00079) 46.4%
Preventive Screens by PCP	0.87	−0.19** (0.0090) −21.7%	Preventive Screens by Specialist	0.27	0.072** (0.0032) 26.3%
			Treated PCP Sample Size	18895	
			Control PCP Sample Size	18895	

Notes: “PCP” stands for primary care physician. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level. Data relies on the main sample shown in Table 1, which only follows patients for one year post-departure. The chronic medication category includes Selective Serotonin Reuptake Inhibitors (SSRIs), antihypertensives, antidiabetics, and statins.

Table A11

Treatment effect of a PCP leaving a clinic controlling for linear time trends.

Type	Mean	Impact	Type	Mean	Impact
Utilization of Clinic Based Services			Aggregate Markers for Poor Care		
Specialist and Primary Care Visits	14.4	−0.21** (0.034) −1.5%	Emergency Department Visits	0.73	0.032** (0.0037) 4.4%
PCP Visits at Other Clinics	1.029	1.4** (0.019) 140.6%	Inpatient Visits	0.38	0.011** (0.0025) 3.0%
PCP Visits at Clinic	3.9	−2.2** (0.023) −55.4%	SNF Visits	0.15	0.0061** (0.0023) 4.1%
PCP Visits at Other Clinics	1.029	1.4** (0.019) 140.6%	Home Health and Hospice Visits	0.30	0.010** (0.0031) 3.4%
Specialist Visits	9.4	0.52** (0.026) 5.5%	Death	0.048	0.0018** (0.00067) 3.7%
Existing SP EM Visits	3.7	0.31** (0.012) 8.3%	Treated PCP Sample Size	19673	
			Control PCP Sample Size	19673	

Notes: “PCP” stands for primary care provider, “SNF” stands for skilled nursing facility. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4 with the addition of linear time trends over the full, matched sample shown in Table 1. The sample contains patients of matched PCPs three years before and one year after exit. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Section 2.2 for other variable definitions. Percent changes divide the treatment effect (“impact”) by the baseline control mean (“mean”). Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

and their patients. The existence of spillovers should indicate if directly affected patients are impacted vis-à-vis staying PCPs, assuming that staying PCPs treat their existing set of patients the same as the patients they inherit from exiting PCPs.

Fig. A7a shows the number of patients seen per month per PCP, grouped by whether the PCP is exiting (red crosses), staying (blue triangles), or new (black circles). I move to the clinic-level, match clinics instead of PCPs, and define time at the relative month t level to observe care patterns at a higher resolution (see Appendix B for more details). Exiting PCPs are defined to exit in $t = 0$. Staying PCPs are observed

continuously at the clinic from $-36 \leq t \leq 24$. New PCPs are any PCP first observed practicing at the clinic from $-36 < t \leq 24$. Zeros are not included in the average, so to aid in the interpretation of Fig. A7a, the total number of PCPs in each clinic is shown in Fig. A7b (the denominator of the average).³⁵

³⁵ The average clinic size is larger than that in the matched PCP sample because solo clinics, which don’t experience spillover effects, are excluded from the clinic-level sample.

Table A12
Additional outcomes.

Type	Mean	Impact	Type	Mean	Impact
Switch to MA	0.043	0.0013 (0.00083) 3.0%	Sampling of Sub-Specialties Nephrologist Visits	0.14	0.024** (0.0028) 17.0%
Preventive Care in Retail Setting Flu Vaccine by Retail	0.19	0.014** (0.0016) 7.6%	Cardiologist Visits	0.93	0.059** (0.0058) 6.3%
Preventive Screens by Retail	0.24	0.075** (0.0053) 31.2%	Gastroenterologists Visits	0.19	0.017** (0.0022) 9.1%
			Surgeon Visits	0.72	0.018** (0.0049) 2.5%
Additional Screens Flu Vaccine	0.49	−0.031** (0.0021) −6.4%	Timing of Visits Mo. Since Visited Any PCP	2.8	0.079** (0.0085) 2.9%
Preventive Screens	1.4	−0.042** (0.0087) −3.0%	Mo. Since Visited Any SP	2.1	−0.044** (0.0044) −2.1%
Annual Exams	0.083	−0.017** (0.0022) −20.9%	Spending Log Tot. Spending	8.0	0.015** (0.0064) —
ED Classification Outcomes ED Visits, Preventable	0.075	0.0020** (0.00094) 2.7%	Log Tot. Out of Pocket	3.8	0.0040 (0.0098) —
ED Visits, Not Preventable	0.20	0.012** (0.0017) 6.2%	Sub Components of Aggregate Definitions Hospice Visits	0.081	0.0068** (0.0024) 8.4%
ED Visits, Primary Care Treatable	0.17	0.0063** (0.0015) 3.7%	Home Health Visits	0.17	0.0045** (0.0022) 2.7%
ED for Non-Emergent	0.25	0.013** (0.0019) 5.0%	Treated PCP Sample Size	19673	
			Control PCP Sample Size	19673	

Notes: “PCP” stands for primary care provider, “ED” stands for emergency department, “Tot” stands for total, “Mo” stands for months, “MA” stands for Medicare Advantage. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4, where patients are followed one year after a PCP’s exit (see Table 1 for baseline balance). All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level.

Fig. A7a shows that staying PCPs see 44.1 more patients per year after a co-working PCPs departure (Table A19).³⁶ However, patients of staying PCPs are not impacted by the exit, suggesting that co-working PCPs are able to compensate for the loss of a co-worker and maintain the same standard of care (Table A19).³⁷ This evidence does not support the hypothesis that patients leave main clinics because they are unable to access care.

6.1.2. Ruling out patients not being able to access care

I further reinforce the findings of Section 6.1.1 by borrowing from tests outlined in Section 6.2 surrounding match quality, facts that are equally relevant here. Fig. A6 and Table 3 show that patients in high PCP density areas—where patients should be more able to access care—have

³⁶ This is likely because clinics do not immediately replace leaving PCPs. Fig. A7b shows that the rate of new PCPs being added to the clinic trends smoothly over time, opposed to suddenly increasing when a PCP exits.

³⁷ “Indirectly affected” patients are patients of staying PCPs and do not ever have a claim billed by exiting PCPs. This analysis assumes that staying PCPs treat their existing set of patients (indirectly affected patients) similarly to the patients they take-on post-departure (directly affected patients).

larger treatment effects than patients in low PCP density areas, providing evidence against an access hypothesis. Table A20 compares patients with and without affiliated nearby clinics—where patients with more similar clinics should have better access—showing similar results. Further, even in clinics that stay open after a PCP’s exit, patients still substitute to specialists and are adversely affected (Table A15). I also redo the match to include whether the patient resides in an urban or rural area, showing that main effects are maintained (Table A17). Taken together, the evidence rules against lack of access being a main mechanism.

6.1.3. Ruling out different PCP practice patterns

A growing literature finds that PCP practice styles explain about 2–3% of the variation in patients’ long-run total utilization (Kwok (2019); Fadlon and Van Parys (2020)). As a result, differences between exiting and replacement PCPs propensity to refer to specialists may explain the sustained decrease in primary care visits and increase in specialty visits.

There are two main pieces of evidence that point against this hypothesis. First, patients who never visit a PCP again drive the aggregate decline in primary care utilization (Table A21). In the commonly used

Table A13

Treatment effect of PCP unexpectedly leaving practice for four years post exit.

Type	Mean	Impact Year 1 Post Exit	Impact Year 2 Post Exit	Impact Year 3 Post Exit	Impact Year 4 Post Exit
Utilization of Services					
Primary Care Visits	5.0	−0.70** (0.025)	−0.73** (0.026)	−0.74** (0.027)	−0.76** (0.028)
Specialist Visits	9.3	−13.9% 0.48** (0.034)	−14.4% 0.38** (0.036)	−14.7% 0.33** (0.039)	−15.0% 0.30** (0.041)
		5.2%	4.0%	3.5%	3.2%
Aggregate Markers for Poor Care					
ED and Inpatient Visits	0.88	0.021** (0.0054)	0.0012 (0.0056)	−0.0029 (0.0058)	−0.00026 (0.0062)
SNF Visits	0.16	2.4% 0.0065** (0.0030)	0.14% 0.0013 (0.0032)	−0.33% 0.0014 (0.0033)	−0.030% 0.0032 (0.0035)
Home Health and Hospice Visits	0.24	3.9% 0.0043 (0.0034)	0.77% 0.0085** (0.0038)	0.85% 0.0059 (0.0041)	2.0% 0.0023 (0.0041)
Death	0.054	1.8% 0.00078 (0.00089)	3.6% −0.00017 (0.00091)	2.5% 0.00014 (0.00094)	0.99% −0.00027 (0.00097)
Treated PCP Sample Size		1.5% 12251	−0.31%	0.26%	−0.49%
Control PCP Sample Size		12251			

Notes: “ED” stands for emergency department, “SNF” stands for skilled nursing facility. Analyses use PCP moves of >100 miles and complete exits using a similar matching strategy as that used in main analyses, but over a sample of PCPs and patients that can be followed four years after exit. The Table displays results from the difference-in-differences specification outlined in Section 4.4, except patients are followed four years after a PCP’s exit. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level. Data relies on a sample that follows patients for four years post-departure.

switcher design, patients who never again visit a PCP will be missed entirely.

Second, I directly test for the importance of this hypothesis by controlling for the specialist and primary care utilization of replacement PCPs’ indirectly affected patients from $1 \leq q \leq 4$ (i.e. the leave-out-mean).³⁸ Treated patients (a) who do not see a new PCP from $1 \leq q \leq 4$ (27% of patients) and (b) with replacement PCPs who only see treated patients over the relevant period (2% of patients) are not included in analyses. Case (a) is excluded because it is unclear what a replacement PCP’s practice patterns look like when a patient does not choose a replacement PCP. Case (b) is excluded because I am unable to quantify the leave-one-out-mean among unaffected patients if the replacement PCP only cares for treated patients. This is a limitation of this method and of the literature more generally.

The magnitude of the coefficients does not significantly change when controlling for the leave-out-mean of specialist and primary care utilization (column 2, Table 4).³⁹ The number of primary care visits declines by 0.73 visits (SE 0.020), which is not significantly different from the decline of 0.71 visits (SE 0.020) in the uncontrolled results. The number of specialist visits is also not significantly different (0.52 vs 0.52 visits). Therefore changing practice styles do not seem to be a main explanation for the long-term decline in primary care visits and increase in specialty visits.

6.1.4. Ruling out alternative mechanisms

There are at least five other reasons for observed effects. First, replacement PCPs may not compensate for the loss of main PCPs because they do not realize they are now responsible for the entirety of a

patient’s primary care. If this was the case, one would expect patients staying at the main clinic to be unaffected because staying, co-working PCPs should be aware that their co-working PCP left. In contrast, Table A15 shows that patients who belong to clinics that stay open after a PCP’s exit still substitute from primary to specialty care.

Second, replacement PCPs may treat new patients differently from their other patients. For example, replacement PCPs may not prioritize new patients. If this was a main explanation, the rate of PCP visits among treated patients should slowly recover over the four years post-departure as the replacement PCP-patient relationship grows. Instead, Fig. A8 shows that patients decrease their use of primary care long-term, with no sign of recovering.

Third, the increase in adverse events post-exit could be due to large returns to primary care. Table A21 helps illuminate this hypothesis by showing that patients who visit replacement PCPs post-exit have a similar number of primary care visits. However, adverse events still increase, suggesting that it’s about the specific exiting PCP rather than primary care more broadly.

Fourth, the results show that patients use more specialists, home health, and hospice care. One explanation may be that downstream outcomes, like increased hospital stays, are driven by specialists and home health agencies being more likely to refer patients to these settings for treatment. While I cannot rule out this mechanism conclusively, several facts suggest this is not a primary mechanism. A large literature shows that providers with higher skill levels, like specialists, are less likely to use downstream resources.⁴⁰ Additionally, patients visit known specialists after a PCP’s exit, so it’s harder to tell a story of resource use discretely changing when a PCP exits, if everything else about a patient’s relationship with a specialist is constant. Lastly, Table A12 shows that much of the increase in hospice and home health visits is driven by an 8.4% increase in patients’ use of hospice, as opposed to a 2.7% increase

³⁸ I control for 100 quantiles of SP and PCP utilization. In order to attribute utilization to a specific PCP, I assign all patients to the modal PCP seen from $1 \leq q \leq 4$, following prior work Kwok (2019). Using this assignment mechanism, I find the median PCP sees 76 non-focal patients over this period.

³⁹ In column three of Table 4, I also add controls for the replacement PCP’s leave-one-out mean mortality rate among patients. I expand on this in Section 6.1.4.

⁴⁰ For instance, Chan and Chen (2022) show that patients seeing nurse practitioners in the ED have higher resource utilization but worse patient outcomes compared to patients seeing physicians in the emergency department.

Table A14

Treatment effect of a PCP leaving a clinic by patient subgroup.

	High Risk		Not High Risk		Func.		No Func.		DE		Not DE		Minority		White	
Primary Care Visits	6.4	-0.91** (0.031) -14.2%		-0.67** (0.019) -15.1%	6.0	-0.91** (0.031) -15.1%	4.6	-0.68** (0.019) -14.8%	5.2	-1.034** (0.042) -19.8%	4.9	-0.69** (0.020) -14.1%	4.9	-0.85** (0.044) -17.5%	5.0	-0.72** (0.020) -14.5%
Specialist Visits	13.3	0.64** (0.051) 4.8%	p < 0.001	0.48** (0.027) 6.0%	12.3	0.65** (0.049) 5.2%	8.3	0.46** (0.027) 5.5%	8.2	0.57** (0.014) 6.9%	9.5	0.50** (0.056) 5.2%	8.1	0.62** (0.028) 7.6%	9.7	0.50** (0.061) 5.1%
ED and Inpatient Visits	1.3	0.046** (0.0098) 3.5%	0.001	0.037** (0.0043) 5.3%	1.3	0.065** (0.0095) 4.8%	0.68	0.027** (0.0042) 4.0%	1.4	0.064** (0.014) 4.4%	0.72	0.035** (0.0041) 4.9%	0.99	0.074** (0.013) 7.5%	0.85	0.037** (0.0044) 4.3%
SNF Visits	0.25	-0.00046 (0.0055) -0.18%	0.39	0.0085** (0.0023) 7.4%	0.25	0.0063 (0.0050) 2.5%	0.11	0.0055** (0.0024) 4.9%	0.24	0.000010 (0.0065) 0.0044%	0.13	0.0066** (0.0024) 4.9%	0.11	-0.00028 (0.0060) -0.24%	0.15	0.0058** (0.0024) 3.8%
Home Health and Hospice Visits	0.50	0.013* (0.0070) 2.5%	0.11	0.011** (0.0032) 5.1%	0.48	0.014** (0.0065) 2.9%	0.22	0.0081** (0.0032) 3.6%	0.41	0.0020 (0.0081) 0.48%	0.27	0.012** (0.0033) 4.6%	0.30	0.014* (0.0080) 4.6%	0.29	0.0099** (0.0034) 3.4%
Death	0.078	0.0043** (0.0016) 5.6%	0.83	0.0014** (0.00067) 4.1%	0.061	0.0038** (0.0014) 6.2%	0.041	0.0013* (0.00072) 3.2%	0.052	0.0024 (0.0017) 4.6%	0.045	0.0017** (0.00071) 3.7%	0.030	0.0042** (0.0017) 13.8%	0.048	0.0014** (0.00072) 3.0%
Treated PCPs	19220		0.09			0.10			0.70			0.13				
Control PCPs	19220		19204	17354	19402	12805			19058	10079		18748				

Notes: "Func" stands for functional limitations, "DE" stands for dual eligible. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays indicator variables from the difference-in-differences specification outlined in Section 4.4. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Comparisons between groups are within PCP using the same regression. Bolded estimates indicate that the groups are significantly different at the 5% level. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level. Data only follows patients for one year post-departure. See Section 2.2 for more details on how heterogeneity and variables were defined.

Table A15

Treatment Effect of a PCP Leaving a clinic that stays open after a PCP's exit.

Type	Mean	Impact	Type	Mean	Impact
Utilization of Clinic Based Services			Aggregate Markers for Poor Care		
Specialist and Primary Care Visits	14.2	−0.17** (0.036)	Emergency Department Visits	0.74	0.024** (0.0041)
Primary Care Visits	4.9	−1.2% −0.55** (0.021)	Inpatient Visits	0.38	3.3% 0.0084** (0.0028)
PCP Visits at Clinic	3.9	−11.1% −1.6** (0.025)	SNF Visits	0.15	2.2% 0.0039 (0.0025)
PCP Visits at Other Clinics	1.0059	−40.0% 1.015** (0.021)	Home Health and Hospice Visits	0.29	2.6% 0.0080** (0.0035)
Specialist Visits	9.3	100.9% 0.38** (0.029)	Death	0.048	2.7% 0.0019** (0.00075)
Existing SP EM Visits	3.7	4.1% 0.23** (0.013)	Treated PCP Sample Size		4.0% 14399
		6.3%	Control PCP Sample Size		19673

Notes: “PCP” stands for primary care provider, “SP” stands for specialist, “EM” stands for evaluation and management, and “SNF” stands for skilled nursing facility. Analyses use PCP moves of >100 miles and complete exits where the home clinic stays open after a PCP's exit in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Table displays results from the difference-in-differences specification outlined in Section 4.4, which follows patients one year post-departure. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Section 2.2 for other variable definitions. Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

Table A16

Treatment effect of a PCP leaving full departures.

Type	Mean	Impact	Type	Mean	Impact
Utilization of Clinic Based Services			Aggregate Markers for Poor Care		
Specialist and Primary Care Visits	14.4	−0.23** (0.040)	Emergency Department Visits	0.72	0.042** (0.0044)
Primary Care Visits	5.0	−1.6% −0.84** (0.024)	Inpatient Visits	0.36	5.8% 0.011** (0.0030)
PCP Visits at Clinic	3.9	−16.9% −2.4** (0.027)	SNF Visits	0.081	2.9% 0.0026* (0.0013)
PCP Visits at Other Clinics	1.026	−60.0% 1.5** (0.023)	Home Health and Hospice Visits	0.21	3.2% 0.0047* (0.0027)
Specialist Visits	9.5	148.4% 0.61** (0.031)	Death	0.047	2.3% 0.0015* (0.00078)
Existing SP EM Visits	3.7	6.4% 0.37** (0.014)	Treated PCP Sample Size		3.2% 13671
		9.8%	Control PCP Sample Size		13671

Notes: Analyses only use PCP moves where the PCP completely exited the data. The Table displays results from the difference-in-differences specification outlined in Section 4.4, which follows patients one year post-departure. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. “SNF” stands for skilled nursing facility. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Section 2.2 for other variable definitions. Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

in patients' use of home health. While home health may be referring patients directly to hospice, given that the increase in hospice utilization is almost fourfold larger, it seems unlikely to be the main story.

Fifth, the increase in adverse events post-exit could be because patients switch to lower quality PCPs. Column three of Table 4 adds controls for the replacement PCP's leave-one-out mean mortality rate among patients. It shows that the rate of ED and inpatient visits is unchanged across columns. This suggests that patients aren't switching to lower quality PCPs, at least in terms of mortality. Further, Section 5.1 also shows that differences between exiting versus replacement PCPs experience, training, and age are small.

6.2. Why do patients value relationships?

A “relationship” refers to repeated interactions between two parties that know each other. This section develops two main hypotheses regarding the theory of relationships and explores which hypothesis more closely matches the evidence. Although not causal, the evidence more closely aligns with a model where relationships result from an “experience” component that increases the value of collaboration over time (Becker, 1962; Polanyi, 1966; Jaravel et al., 2018).

While suggestive, results are not conclusive. For instance, the true model may be a mix of the two or even a distinct hypothesis.

Table A17

Treatment effect of a PCP leaving a clinic including urban indicator in match.

Type	Mean	Impact	Type	Mean	Impact
Utilization of Clinic Based Services			Aggregate Markers for Poor Care		
Specialist and Primary Care Visits	14.5	−0.16** (0.034)	Emergency Department Visits	0.73	0.032** (0.0038)
Primary Care Visits	5.0	−1.1% −0.70** (0.020)	Inpatient Visits	0.38	4.3% 0.0067** (0.0026)
PCP Visits at Clinic	3.9	−14.1% −2.2** (0.023)	SNF Visits	0.16	1.8% 0.0043* (0.0024)
PCP Visits at Other Clinics	1.035	−55.0% 1.5** (0.020)	Home Health and Hospice Visits	0.30	2.7% 0.011** (0.0032)
Specialist Visits	9.5	140.8% 0.54** (0.027)	Death	0.048	3.6% 0.0017** (0.00068)
Existing SP EM Visits	3.8	5.7% 0.32** (0.012)	Treated PCP Sample Size	18895	3.6%
		8.5%	Control PCP Sample Size	18895	

Notes: Analyses use PCP moves of >100 miles and complete exits using a match similar to the main matching strategy, but with the addition of an urban/rural indicator that should control for differences in the local availability of providers. The Table displays results from the difference-in-differences specification outlined in Section 4.4, which follows patients one year post-departure. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. “SNF” stands for skilled nursing facility. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Section 2.2 for other variable definitions. Percent changes divide the treatment effect (“impact”) by the baseline control mean (“mean”). Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

Table A18Treatment effect of a PCP Moving ≤ 40 minutes away.

Type	Mean	Impact	Type	Mean	Impact
Utilization of Clinic Based Services			Aggregate Markers for Poor Care		
Specialist and Primary Care Visits	14.9	−0.30** (0.066)	Emergency Department Visits	0.68	0.0085 (0.0067)
Primary Care Visits	5.2	−2.0% −0.34** (0.033)	Inpatient Visits	0.39	1.3% 0.0062 (0.0048)
PCP Visits at Other Clinics	0.98	−6.6% 2.7** (0.040)	SNF Visits	0.083	1.6% −0.00099 (0.0022)
Prob. Switch to Original PCP at Diff Clinic	0.023	273.8% 0.14** (0.0035)	Home Health and Hospice Visits	0.23	−1.2% 0.0057 (0.0052)
Specialist Visits	9.7	576.5% 0.039 (0.052)	Death	0.049	2.5% 0.0014 (0.0012)
Existing SP EM Visits	3.8	0.40% 0.022 (0.021)	Treated PCP Sample Size	4683	2.9%
		0.56%	Control PCP Sample Size	4683	

Notes: Analyses use PCP moves of ≤ 40 min and match the treated sample to a similar control sample following the main matching strategy outlined in Section 4.2. The Table displays results from the difference-in-differences specification outlined in Section 4.4. The sample contains patients of matched PCPs three years before and one year after exit. All regressions are estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. “Prob.” indicates that the outcome is the yearly probability. “SNF” stands for skilled nursing facility. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. See Section 2.2 for other variable definitions. Percent changes divide the treatment effect (“impact”) by the baseline control mean (“mean”). Family wise p-values are denoted by asterisks as follows: * $p < 0.10$ ** $p < 0.05$.

Nevertheless, by presenting results in the framework conceptualized below, the hope is to illuminate new facts and suggest fruitful areas for future work.

Conceptualization. I apply the concepts of job matching and firm-specific capital between employees and firms to relationships between PCPs and patients.⁴¹ Matching, or match quality, can be interpreted as patients paying a fixed, time-invariant cost to search for a PCP that is a

“good match” (Jovanovic, 1979a). Alternatively, good relationships may be created over time. For instance, relationships may be valuable because of the experience, friendship, and trust specific to that particular PCP-patient relationship, information that takes time to build (Becker, 1962; Polanyi, 1966; Jovanovic, 1979b).⁴²

Testing for Match Quality. The main theoretical implication of

⁴¹ Jaravel et al. (2018) make the same application to inventors.

⁴² This is similar to Jovanovic (1979b)’s theory of firms as experience goods and Polanyi (1966)’s concept of tacit knowledge.

Table A19

Treatment effect of PCP unexpectedly leaving practice on clinic-level outcomes.

Type	Mean	Impact Year 1 Post Exit	Impact Year 2 Post Exit	P-Value
Firm Level Outcomes				
Avg Number of Pat Seen Per Month-PCP, Staying PCPs	214.9	44.1*** (6.0) 20.5%	38.1*** (6.5) 17.7%	0.02
Avg Number of Pat Seen Per Month-PCP, New PCPs	23.9	35.7*** (6.1) 149.5%	38.7*** (6.5) 162.1%	0.04
Count of New PCPs	0.056	0.028*** (0.0064) 51.2%	0.018*** (0.0061) 33.2%	0.10
Treated Clinic Sample Size	1573			
Control Clinic Sample Size	1573			
Indirectly Affected Patients' Outcomes				
Utilization of Clinic Based Services				
Primary Care Visits	5.0	-0.050 (0.046) -1.0080%	-0.0012 (0.055) -0.025%	0.11
Specialist Visits	9.8	-0.030 (0.060) -0.31%	0.091 (0.071) 0.92%	0.02
Urgent Care Visits	0.015	-0.00061 (0.0012) -4.2%	0.00072 (0.0011) 5.0%	0.21
Aggregate Markers for Poor Care				
ED and Inpatient Visits	0.89	0.0033 (0.0091) 0.37%	0.012 (0.0095) 1.3%	0.33
Treated Clinics	1558			
Control Clinics	1558			

Notes: Analyses use PCP moves of >100 miles and complete exits after a PCP's exit in a matching clinic-level sample. The Table displays results from a specification similar to the difference-in-differences specification outlined in Section 4.4, with one large difference: analyses rely on a data set that matches clinics, not PCPs, and follows patients two years post-departure. As a result of the clinic-level match, clinics with only one PCP are not included. Regressions are at the clinic-year level, contain clinic fixed effects, and cluster at the clinic-level. Indirect patients are patients who were never observed to visit a departing PCP. Regressions are at the clinic-year level, contain clinic fixed effects, and cluster at the clinic-level. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level. See Section 2.2 for variable definitions and Appendix B for more details on how the clinic-level sample was created.

Table A20

Treatment effect of a PCP leaving a clinic focused on clinics that stay open after a PCP's exit. With More Than One Similar Clinic Available

Type	Mean	Similar TIN Surrounding Clinic		P-Value
		No Similar Clinic	At Least One Similar Clinic	
Primary Care Visits	4.9	Impact -0.67** (0.072) -13.8%	Impact -0.54** (0.022) -11.1%	0.08
New PCP Relationships	0.14	0.12** (0.0084) 89.1%	0.087** (0.0028) 71.9%	p < 0.001
Specialist Visits	9.9	0.45** (0.080) 4.6%	0.37** (0.031) 4.0%	0.34
Treated PCPs	102		14297	
Control PCPs	19		19654	

Notes: Analyses use PCP moves of >100 miles and complete exits after a PCP's exit in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The table displays results from the difference-in-differences specification outlined in Section 4.4, but focused on PCP departures where the clinic stays open after a PCP's exit. All regressions are estimated relative to year = -2 to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Data relies on the main sample shown in Table 1, which only follows patients for one year post-departure. "Similar clinics" have the same tax identifier as the patient's main clinic, which indicates the same management. The average clinic does not have a similar surrounding clinic, so I compare clinics with zero to those with at least one similar clinic. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level.

search-and-matching models is that, if relationships are found (and not made), a new PCP of similar quality should perfectly replace the exiting PCP and patients should return to baseline utilization after re-matching.

Empirically, this implies that patients ability to rematch to a new PCP should be independent of the characteristics of their prior match.⁴³

I test for the importance of match quality in three ways. First, if

⁴³ In this case, the length of a patient's relationship with a PCP should not be predictive of outcomes.

Table A21

Treatment effect of a PCP leaving a clinic by whether the patient visits a replacement PCP.

	Replacement PCP		No PCP		
Type	Mean	Impact	Mean	Impact	P-Value
Primary Care Visits	5.4	0.025 (0.018) 0.46%	3.2	-3.3** (0.024) -103.0%	p < 0.001
Specialist Visits	10.0	0.85** (0.027) 8.5%	7.4	-0.61** (0.040) -8.2%	p < 0.001
Home Health and Hospice Visits	0.23	-0.0065** (0.0025) -2.9%	0.20	0.056** (0.0034) 28.2%	p < 0.001
ED and Inpatient Visits	0.95	0.048** (0.0045) 5.1%	0.76	0.018** (0.0062) 2.4%	p < 0.001
Treated PCPs	19468		17783		
Control PCPs	19468		17783		

Notes: The Table displays results from a single regression, difference-in-differences specification outlined in Section 4.4, where the interaction terms are reported in the Table. The regression is estimated relative to $year = -2$ to allow for anticipatory effects without biasing main estimates. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Data relies on the main sample shown in Table 1, which only follows patients for one year post-departure. “Replacement PCP” indicates that the patient was observed visiting a PCP (either an exiting or new PCP) at least once in the 12 months following their main PCP’s exit. “No PCP” indicates that the individual was not observed visiting any PCP after patients’ main PCPs’ exit. Stars indicate significance at the 10% (*), 5% (**), and 1% (***) level.

patients search for a PCP that is a similarly good match, we would expect to see patients trying new PCPs. In contrast, only 23% of treated patients visit a new PCP and, of patients who do, they do not visit multiple new PCPs, instead visiting one new PCP on average.⁴⁴

Second, I follow the classical test for match quality and quantify if patients in thicker PCP markets are more likely to rematch (Lazear, 2009). In thicker markets, patients should be more able to find a similarly good replacement PCP and therefore be less impacted by the loss of a PCP. Table 3 shows treatment effects in high versus low density areas. The primary outcome of interest is the rate of forming a new relationship with a replacement PCP. If match quality drives effects we would expect patients in high density areas to be *more* likely to form new PCP relationships, regaining their prior level of utilization once they rematch. The Table instead shows that patients are equally likely to form new PCP relationships in high opposed to low density areas (See Fig. A6 for event plots).

Third, I move beyond the standard definition of market thickness used in the literature and consider cases where patients are surrounded by clinics with similar ownership structures. Patients may be better able to rematch to a PCP of similar quality across affiliated clinics, which likely share health records and staff. To test this, I compare patients who lose a PCP with and without a nearby affiliated clinic. I find evidence against this mechanism by showing that patients without a nearby affiliated clinic are slightly *more* likely to form a new PCP relationship (Table A20). Taken together, the empirical evidence does not obviously align with the match quality view of relationships.

Testing for the Experience View of Relationships. The testable implication of relationships growing over time is that treatment effects should be larger among patients with longer relationships with exiting

⁴⁴ To take this a step further, one may be worried that 23% of patients visiting a new PCP reflects restricted availability of PCPs. The second test for match quality—heterogeneity by market thickness—suggests this is not a main story by showing that patients in thicker PCP markets are not more likely to make a new PCP relationships. Further, Section 6.1.1 quantifies the impact of a PCP’s exit at the main clinic, showing that staying PCPs at affected clinics take on more patients after a PCP’s exit with no measurable impact on patient care. This suggests that patients can continue to seek care at main clinics, but instead choose not to, pushing against the restricted availability hypothesis.

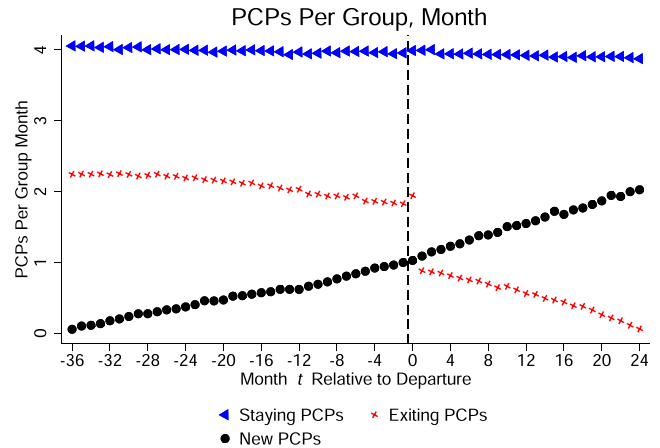


Fig. A2. PCPs per group over time. Notes: Graphs rely on a sample that matches clinics, instead of PCPs, which is described in Appendix B. Fig. A7b shows the number of PCPs per group over time. The average number of exiting PCPs does not go to zero at $q = 1$, instead showing that there are subsequent departures at the group. PCPs that exit after the focal PCP departure in $q = 0$ gradually exit over the two year post period.

PCPs. I test this by comparing treatment effects across the length of patients relationship with exiting PCPs.⁴⁵

All analyses use PCP fixed effects to compare patients with the same exiting PCP to control for PCP and clinic-level factors that may differ across patients, like exiting PCP quality and the local availability of PCPs. I also exclude patients under age 75 (as of $q = -4$) to circumvent left censoring due to observing most patients when they become Medicare eligible at age 65. Table A7 shows that these restrictions leave a well balanced sample. Patients are aged similarly across bins and, if anything, patients with shorter relationships have observably worse health than patients with longer relationships. Further, Fig. A4 shows that pre-trends are minimal.

Fig. 5 shows that treatment effects monotonically increase across relationship bins. Patients with longer relationships are more likely to

⁴⁵ See Section 2.2 for more details on how this sample was defined. See Appendix C for comparisons across subgroups at baseline.

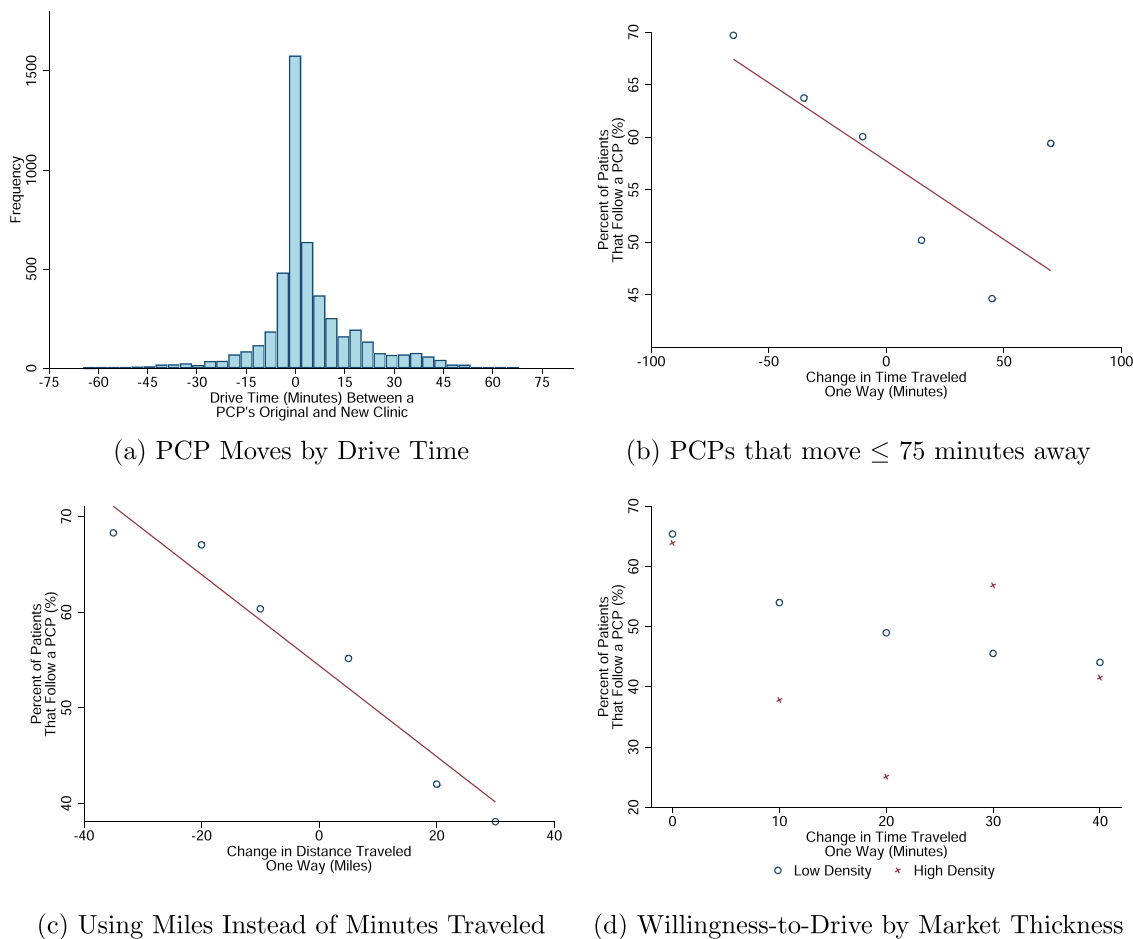


Fig. A3. Percent of patients following a moving PCP. *Notes:* The binscatters show the percent of patients that follow a PCP by the distance the PCP moves from the original clinic, relative to the patient's zip code of residence. Negative distances indicate that the PCP moved closer to the patient's zip code of residence and positive distances show that the PCP moved farther. These figures rely on a different sample than the main analyses. Fig. A3b uses a sample of patients where PCPs moved ≤ 75 minutes away. Fig. A3c shows the percent of patients following using miles traveled, opposed to minutes. Fig. A3d compares low and high density areas defined using the local density of PCPs at the original clinic's zip code.

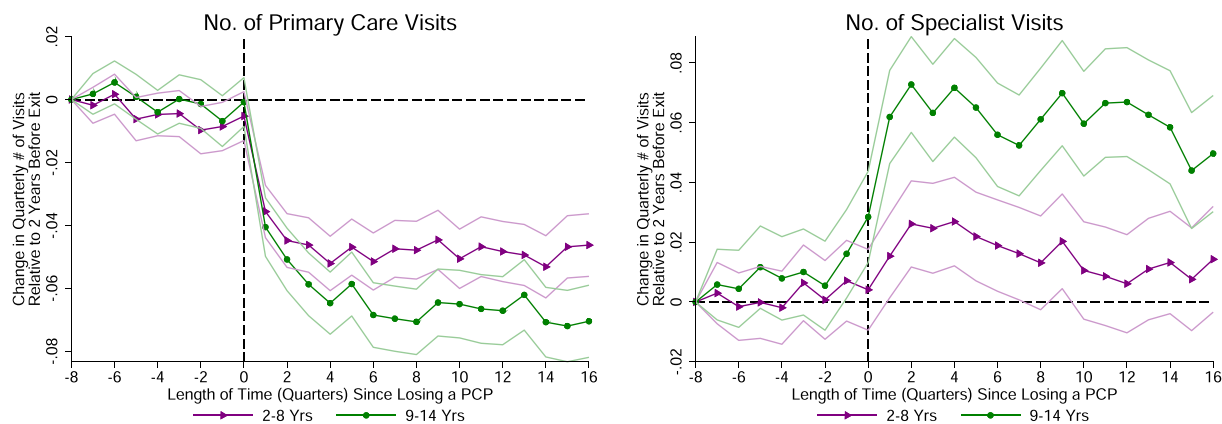


Fig. A4. Treatment effect of a PCP leaving a clinic by relationship length. *Notes:* "PCP" stands for primary care physician. "Yrs" stands for years. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Figure displays results from interaction terms based on a difference-in-differences regression outlined in Section 4.4. The regression is estimated at the PCP-year level, contains pre-departure PCP fixed effects, clusters at the PCP-match level, and is estimated relative to year = -2 to allow for anticipatory effects without biasing main estimates. Patients aged < 75 as of $q = -12$ were dropped from analyses to avoid censoring due to only observing patients once they become Medicare eligible (typically at age 65). The purple markers represent PCP-patient relationships 2-8 years long, whereas the green markers represent relationships 9-14 years long. See Table A7 for balance across groups.

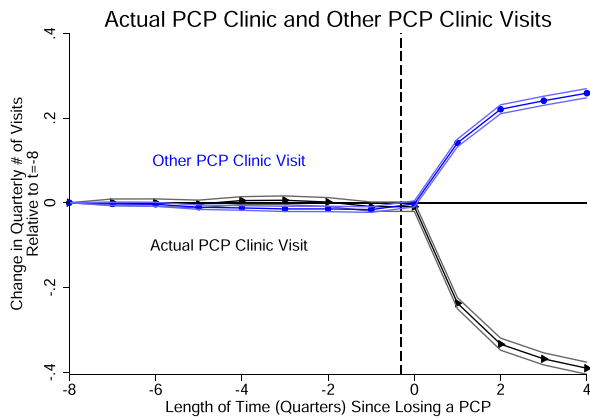


Fig. A5. Impact of a PCP's exit on patients' likelihood of substituting right-arrow a new primary care clinic when main PCP Clinic Stays Open. *Notes:* The event study shows shows patients likelihood of substituting from their original, "actual" PCP clinic to a PCP at a different clinic.

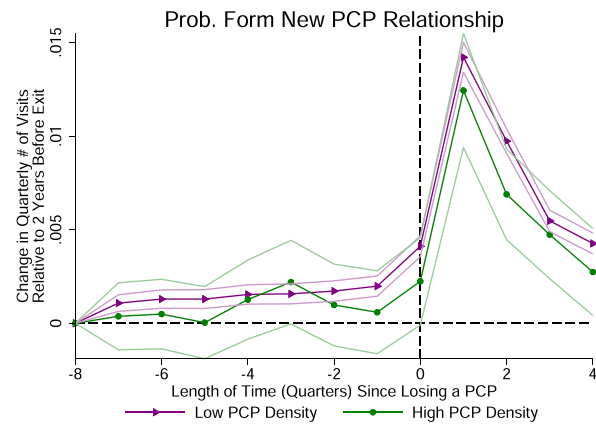
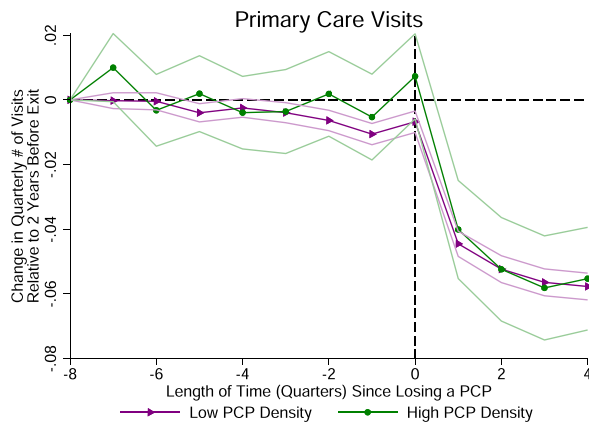
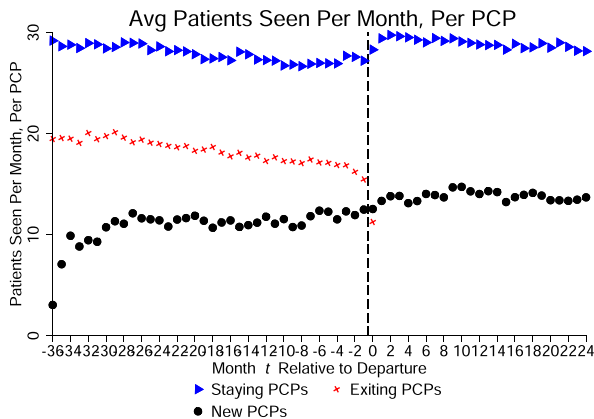
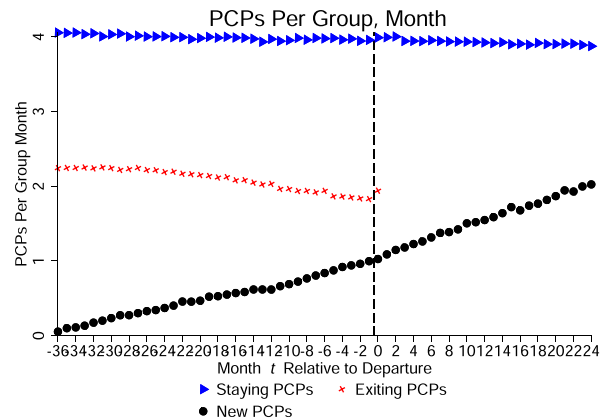


Fig. A6. Treatment effect of a PCP leaving a clinic by local density of PCPs. *Notes:* "PCP" stands for primary care physician. Analyses use PCP moves of >100 miles and complete exits in the matched PCP sample outlined in Section 4.2, with balance shown in Table 1. The Figure displays results from the difference-in-differences specification outlined in Section 4.4, which follows patients one year post-departure. The local density of PCPs is defined as the number of PCPs filing billing claims within a 30 mile radius of each focal clinic ZIP divided by the population. Thick markets are defined to be above average and thin markets are below average density areas. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, are estimated to relative quarter = -8 and cluster at the PCP-match level. Table 3 shows the pooled point estimates.



(a) Average Number of Patients Seen Per Month, Per PCP by Group



(b) Number of PCPs by Type of PCP Over Time

Fig. A7. Clinic-level effects. *Notes:* Graphs rely on a sample that matches clinics, instead of PCPs. This sample does not include practices with one PCP. Additional details are described in Appendix B. Fig. A7a shows the number of patients seen per month, per PCP by type of PCP. Blue triangles represent staying PCPs, a group that includes PCPs that existed at the clinic in $q = -12$. Red crosses represent exiting PCPs, who exit from $0 \leq t \leq 24$. Black circles represent PCPs that are new to the clinic between $-35 \leq t \leq 24$. Because points in Fig. A7a do not account for the size of the group, Fig. A7b shows the number of PCPs per group over time (the denominator of the rate showed in Fig. A7a).

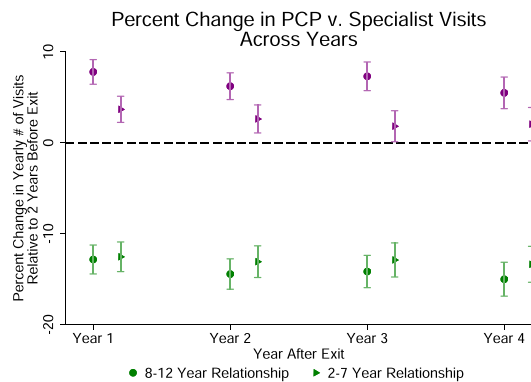


Fig. A8. Effects of a PCP leaving on PCP visits heterogeneity by relationship length. *Notes:* The purple markers above the $y = 0$ axis represent the change in specialist visits and the green markers below the $y = 0$ axis represent the change in PCP visits. Analyses use PCP moves of >100 miles and complete exits after a PCP's exit in the matched PCP sample outlined in Section 4.2, except over a sample that follows patients four years post-exit. The two relationship definitions were defined as above. v. below the mean relationship length. Section 4.4 outlines the underlying specification, but here we follow patients four years. Regressions are at the PCP-year level, contain pre-departure PCP fixed effects, and cluster at the PCP-match level. Patients aged < 75 as of $q = -12$ were dropped from analyses to avoid censoring due to only observing patients once they become Medicare eligible (typically at age 65).

relationship could be endogenous to match quality or preferences. Further, patients with longer relationships may have worse outside options, e.g., if they invest in outside relationships less than patients with shorter relationships before their main PCP's exit or their outside option is lower quality.⁴⁶

7. Conclusion

I establish that patients value relationships and that relationships positively impact patients' health. Patients overwhelmingly do not make new PCP relationships, suggesting that understanding how relationships form is a fruitful area for future work. These findings may be magnified in the Medicare context because Medicare patients are older and in worse health. They also may be muted because Medicare patients face

Appendix A. Additional data details

See Table A1.

A.1. Additional definitions

Each PCP is assigned a unique identifier combining their NPI and associated clinic identifier (NPI-TIN-ZIP). NPIs belonging to organizations, and not individuals, are dropped from the sample because it is not possible to observe individual provider exits. See Section 2 for how I identified whether a provider's NPI belonged to a PCP or specialist.

Clinic Closures. I define clinic closures to occur when a PCP departs and (i) the TIN is the last TIN observed at the ZIP,⁴⁸ (ii) all PCPs at the clinic completely disappear from the data, or (iii) the TIN disappears from the data in month $t + 1$ and the number of NPIs at that nine digit zip decreases by the exact number of NPIs affiliated with that clinic as of month t .

Utilization. "Office settings" are tagged using place of service codes equal to 11 in the carrier file. Medicare Provider Analysis and Review files identify hospitalizations. Total spending follows recent literature and aggregates patients' carrier, inpatient, outpatient, urgent care, and ED charges (Finkelstein et al., 2016). Out-of-pocket costs aggregate the coinsurance and deductibles paid by patients for these services. Prescription drug claims are obtained from the Part D Event and Plan Characteristic files. Medicare Beneficiary Summary files provide patients' date of death, demographic characteristics, and enrollment information. All files are linked using beneficiary identification numbers and claim dates.

low and relatively constant out-of-pocket costs across providers and have large teams of PCPs and specialists. It is unclear how results would generalize to younger patients who may face network constraints, need a PCP for referrals, and have less attachment to individual PCPs. For instance, requiring patients to receive a referral to a specialist may encourage patients to return to primary care and establish new relationships.

Findings support the theoretical role of the PCP as a point of contact to the health system (Starfield, 1994). This role is receiving more focus as recent and proposed policy reforms directly target the PCP-patient relationship. For instance, Medicare's Shared Savings Program shares savings with PCPs who keep total costs below a financial benchmark, encouraging PCPs to take greater control over the continuum of patient care (McWilliams et al., 2016).

PCP departures are especially ubiquitous in the healthcare context where volatile insurance networks and non-compete agreements may artificially sever a PCP-patient relationship, even if the PCP remains in practice. For instance, 26% of commercially insured individuals switch insurance plans in a given year and 39 states enforce non-compete agreements (Barnett et al. (2017) and Hausman and Lavetti (2021)). PCP retirements are also projected to increase over the next decade, with 32% of PCPs currently over age 60 (Sabety et al., 2021). Further, patients likely face frictions when establishing care with a new provider, making the loss of a PCP even more costly (Sabety et al., 2023; Sood et al., 2021).

While policymakers and firms cannot stop PCPs from retiring, they could create exceptions for patients who want to maintain their long-term PCP when switching insurance plans or if a PCP moves across firms and a non-compete agreement is in effect.⁴⁷ Interventions like these may limit the negative impact of severing longstanding PCP-patient relationships, improving patient welfare long-term.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

⁴⁶ Although Table 4 suggests the quality channel is less important by showing that replacement PCPs patients' have similar mortality rates to exiting PCPs.

⁴⁷ This directly relates to recent legislation proposing to ban non-compete clauses.

⁴⁸ This would not include clinics that switched TINs due to an ownership change because I would observe a new clinic at that ZIP in that case.

Medicare Advantage Patients. Patients were coded as being in Medicare Advantage (MA) according to the Master Beneficiary Summary file.

Definition of New v. Existing Relationships. Patients classified as “new patients” through evaluation and management codes did not receive any professional services from the provider or another provider of the same specialty in the same group practice within the previous 3 years. Existing patients are individuals who received care from providers of the same specialty in the same group practice within the previous three years.⁴⁹

Additional Patient Sample Restrictions. Patients who switched in and out of being enrolled in MA were dropped from the sample to avoid missing data on the patient while they were in MA. All remaining patients who enrolled in MA from $-12 \leq \text{quarters} < -8$, or over the assignment period, were dropped from the main sample.⁵⁰ Patients that died over the assignment period were also dropped from the sample. This was done to precisely define a PCP’s pool of patients from $-12 \leq \text{quarters} < -8$.

A.2. Additional description of departures

See Fig. A1 and Tables A2,A3,A4.

Appendix B. Clinic-level sample

Treated Clinics: The same departure definition was used as in Section 2.2. Treated clinics (i) existed for 49 months, (ii) see an average of ≥ 2 patients per month over the period, and (iii) see ≥ 30 patients from $-12 \leq q \leq -8$. This algorithm drops clinic closures and solo clinics. Further, clinics with > 100 PCPs were excluded.

Control Clinics: If a PCP departed a clinic but the conditions laid out in Section 2.2 were not met, the PCP would be considered a control and, as such, the PCP’s clinic would be called a “control clinic.” Further, control clinics had to (i) exist for 49 months, (ii) see an average of ≥ 2 patients per month over the period, and (iii) see ≥ 30 patients from $-12 \leq q \leq -8$.

Matching: Three coarsened bins of average PCP age at the clinic, seven coarsened bins of the number of PCPs per practice, nine coarsened bins of the number of patients seen in $q = -12$, whether or not the clinic was on an individual or shared model, and month and year of calendar time were matched on.

Indirectly Affected Patients: I call a patient indirectly affected if they *never* saw a departing PCP. I assign patients to clinics in $-12 \leq q \leq -8$ and then I assign patients to PCPs within the clinic based on their modal PCP.

Additional Sample Restrictions: Clinics where the total number of PCPs changed by more than 2 standard deviations in a given year were dropped. This additional restriction was imposed on the clinic sample because being able to accurately draw clinic barriers is more important for clinic-level analyses, opposed to PCP level analyses. This drops 30% of treated clinics. Analyses without this restriction were run for robustness and showed that the number of visits at non-focal clinics decreased in response to a focal-PCP departure, which is unintuitive and suggests that clinic boundaries were likely incorrect in this case.

Appendix C. Balance across subgroups

Tables A5,A6,A7,A8,A9

Appendix D. Additional results

Tables A10,A11,A12,A13,A14,A15,A16,A17,A18,A19,A20,A21.

Appendix E. Additional figures

Figs. A2,A3,A4,A5,A6,A7,A8.

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⁴⁹ CMS’ coding rules can be found here: <https://www.cms.gov/Outreach-and-Education/Medicare-Learning-Network-MLN/MLNProducts/Downloads/eval-mgmt-serv-guide-ICN006764.pdf>

⁵⁰ They were dropped once it was determined that patients do not differentially switch to MA at the threshold. This is checked and quantified in Table A12.

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